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Climate change education through gamification: Escape room as an educational tool to assess pre-service teachers' reflection and consensus decision-making.

Juliana Friedrichsen & Elisa Lehtonen

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Philosophical Faculty</td><td colspan="4">Osasto – School

School of Applied Educational Science and Teacher Education</td></tr>
<tr><td colspan="6">Tekijät – Author

Juliana Friedrichsen & Elisa Lehtonen</td></tr>
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Climate change has become an important topic to research due to its complexity and multidimensional impacts. Education is seen as a key component in tackling climate change issues as it can develop knowledge and the necessary skills to enact action for mitigation and adaptation. In turn, educators are a fundamental part of this transformative process, whose professional practices require specific skills. In this context, it is necessary to assess educators' capability to reflect and make decisions around climate change education. This qualitative study proposed the use of a gamified learning space with an escape room set up to assess how pre-service teachers could reflect and make decisions over climate change issues. Pre-service teachers attending to course Education for Sustainable Future were allowed to voluntarily participate in the research. Five groups took part in the research and each group consisted of four to five participants. First, the data was collected through video recordings and observations from the escape game. Secondly, participants were interviewed together as a group. Then few participants were selected for an individual interview. The research utilized a reflection assessment model to identify whether participants were able to describe the experience properly by describing essential elements, being critical, and drawing conclusions. Results showed pre-service teachers presenting various levels of reflection during the game, and consensus decision-making was an easy-to-reach process. External factors such as language skills, environment, and peer pressure influenced reflection quality. Overall, the escape room demonstrated to be a positive learning experience for pre-service teachers.</td></tr>
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Abbreviations

CCE	Climate change education
COP27	27th Conference of the Parties
ESD	Education for Sustainable Development
IPCC	Intergovernmental Panel on Climate Change
UN	United Nations
UNFCCC	The United Nations Framework Convention on Climate Change

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1. INTRODUCTION

Climate change has become an important topic to research since it has globally large environmental, social, and economic consequences (Ratinen, 2011). We need research and education to understand what actions are required and to find solutions to mitigate and adapt to climate change. In this context, climate change education is seen as a fundamental component of the school curriculum due to its transformative potential (UNESCO, 2020). However, providing meaningful climate change education for students requires high quality education for educators (Boon, 2016). Understanding climate change and environmental issues crucially affects how much one is willing to change one's own habits and it is reported that pre-service teachers in Finland have challenges in understanding science and climate change (Ratinen, 2011). Therefore, there is a need to assess and understand the abilities of these future professionals, regarding not only their knowledge of climate change, but also their skills and pedagogies to promote such education.

Consensus decision-making can be challenging, especially around climate change, since the difference of individuals beliefs and values and the chance of manipulation can influence the consensus decision-making process (Goldberg et al., 2020; Sartor & Brown, 2004). But when successfully made, it can create meaningful experiences and develop cooperation and reflection skills (Sartor & Brown, 2004). When discussing climate change and sustainability issues all aspects should be considered and the decisions made to mitigate climate change require thought through decisions. Consensus is an essential matter when making these decisions. It requires participation from different parties to ensure that multiple perspectives are included in the process to reach a common goal (Bressen, 2012). In education, consensus can be applied in various situations, which can be beneficial for both teachers and students. Consensus can enrich learning experiences and promote a more constructive learning environment (Sartor & Brown 2004). Additionally, teachers must possess skills to make decisions with others and constantly reflect on their work to evolve and develop. Thus, we made the conscious decision to involve pre-service teachers in this research, which can benefit them for the future.

Reflection has also been perceived as a challenging skill and method to apply in practice (Rogers, 2001). However, it has been found to be a crucial skill to possess in the modern world. Teachers should be able to reflect and promote reflection continuously in their work. Reflection has also been highlighted in the Finnish core curriculum, since it improves development, learning and various important skills, such as problem-solving, critical thinking and interpersonal skills (Eronen & Toikka, 2021; Opetushallitus, 2014). Such discoveries point out the benefit of this skill.

Gamification is the utilization of game features into different environments. It is a well-known strategy due to its capacity to bring people together to collaborate, increasing engagement and participation (Huotari & Hamari, 2012). In education gamification has been a growing trend in recent years, being implemented as an innovative pedagogy, especially because it dialogues with the new generations of learners and their demands. Game-based learning has been shown to be effective in the development of crucial skills while gaining content knowledge (Strickland and Kaylor, 2016). Considering the positive impacts of this learning methodology, gamification can be used as a strategy to discuss delicate topics, such as climate change. The experiential aspect of gamification may be an evoker for reflection. In addition, with the proper approaches and design, gamified education can promote decision-making situations, in which participants need to negotiate their values and beliefs to reach consensus.

In this experimental study we propose the use of a gamified learning space with an escape room set up as a tool to promote discussion about climate change among pre-service teachers. We aimed to assess group reflection and consensus decision-making while debating climate change issues. We perceive reflection and consensus decision-making fundamental skills for teachers to create sustainable education for future generations. Thus, we wanted to research how pre-service teachers use these skills in order to discuss and make meaningful decisions together. Moreover, it was essential to find out how knowledge and other factors influence their reflection and decision-making skills.

2. THEORETICAL FRAMEWORK

As mentioned, reflection and consensus decision-making are stated to be challenging skills, but integral parts of solving problems, especially complex issues such as climate change. Therefore, reflection and consensus decision-making processes will be elaborated next. Afterwards, the theory of climate change will be discussed, and the importance and benefits of climate change education will be highlighted. Lastly, it will be considered how gamification can strengthen climate change education to improve reflection and consensus decision-making skills.

2.1 Reflection

Reflection can be perceived as a natural learning process, of thinking thoroughly about a particular matter (Moon, 1999). The starting point of a reflective process is experience (Koole et al., 2011). Moreover, individuals should be open, willing, and engaged to be able to make the reflective process happen (Rogers, 2001). According to Brubacher et al. (1994) when learning about social problems such as environmental issues, formal and direct instruction is not seen as sufficient way to learn. If the goal of learning is to change attitudes and behaviors students should be given mature opportunities to critical thinking, discussion, exploration, problem solving and presentation. Learning is more likely to be effective when students are given the chance to engage in action where they can discuss values and issues involved in their lives.

Reflection is an integral part of teaching. Teachers should constantly think critically about teaching and evaluate their work experiences to improve student learning as well as to provide effective teaching and learning outcomes. It is believed that children are often curious about their surrounding world (Brubacher et al., 1994). Therefore, it is teachers' duty to support and utilize their curiosity for children to learn how they can interact and impact the world. Defective reflection can prevent individuals from questioning their behavior and actions (Koole et al., 2011). Thus, it is essential to provide opportunities for future teachers to reflect and realize the importance of adjusting teaching and responding to issues. In the end, the purpose of reflection is to “integrate understanding gained into one’s experience in order to enable better choices or actions in the future” (Rogers, 2001, p.41), which this research is hoping to achieve additionally as a secondary goal.

Brubacher et al. (1994) describe teachers as reflective practitioners and decision makers. In fact, reflection and decision-making process are perceived being connected with each other. Teachers

should be rational, consider different alternatives and criteria to make decisions. Therefore, it is important to stimulate teachers reflective thinking and reflection skills for them to be able to justify their decisions. Considering the above, it is essential for pre-service teachers to develop reflection skills and practice problem solving to become competent professionals and independent learners.

Reflection has traditionally been understood as an individual process, but social interaction is receiving more and more attention and expected to improve reflection (Koole et al., 2011). Rogers (2001) describes a few techniques, which include interaction with other individuals to foster reflection. Structured experiences in a group and exploring multiple perspectives in a challenging situation are two examples, which facilitate reflection and enhance professional effectiveness. Moreover, an encouraging environment where the facilitator or other group members are asking thought-provoking questions and offering an opportunity to consider perspectives and solutions can improve reflection. It is also crucial for the environment to be as safe as possible to fully take advantage of individuals reflection (Koole et al., 2011).

Internal factors may affect the reflection process. Motivation, expectancy, and prior experiences with reflection might interfere in the entire process. If previous experiences of reflections are positive, it will most likely influence positively into one's own motivation to reflect (Koole et al., 2011). However, Rogers (2001) mentioned that reflection can only be developed by challenging individuals to reflect. Students too often expect learning experiences to be easy and simple, which is a possibility, but does not necessarily promote development. He adds that at times, reflection should be uncomfortable, so one is able to learn the skill. Reflective practices require both support and challenge to make it effective. Building on that, Eyler et al. (1996) found reflective experiences to be meaningful and effective when they are continuous, connected, challenging, and contextualized. Continuous means ongoing reflection, which happens before, during and after an experience and is continued along learner's education. Connected reflection integrates studying content, such as theory, and service experiences together. Meaning a learner can utilize knowledge during an experience. Challenging reflection encourages a learner to think more critically and to question events and issues. Conceptualized reflection means when reflection is implemented in a meaningful context and a setting.

According to Rogers (2001) reflective practices should be trained both during action and after the experience. He also highlighted that reflection can be facilitated and fostered in multiple ways, by reflecting individually, with peers or with a tutor. Although some authors suggest that reflection is more effective when one reflects on one's own (Seibert & Daudelin, 1999), some techniques can be

applied to improve reflection quality in groups. It is suggested to be beneficial to use a facilitator in a group setting to guide the reflection process. It is to make sure that each participant achieves the intended goal from the experience and can think beyond the problem they are working with (Rogers, 2001; Eyler et al., 1996). In this case, the reflective experience can become powerful tool to support individuals understand different perspectives and thoughts (Eyler et al., 1996)

Ash and Clayton (2004) explain that reflection has been perceived by educators as a challenging concept to apply in practice. The lack of structures complicates the instructor's know-how to guide students in the reflection process. However, it is known that complex challenges have a high correlation with problem-solving skills development (Beghetto, 2018, Eronen & Toikka, 2021). Such challenges stimulate pupils to think creatively and critically (Beghetto, 2018). Additionally, challenges seem to raise emotions that are fundamental to the construction of reflection over a problem, for example in mathematical problems (Eronen & Toikka, 2021). As there are still obstacles on the implementation and assessment of reflection, it is up to educators or moderators to develop best practices to facilitate the reflection process. Instructors must provide clear and prompt questions to individuals for them to be able to understand and articulate their learning, thus producing quality reflection. It is essential to encourage individuals to justify their answers to avoid obviousness, inviting them to build up their thoughts, in a conscious process of revisiting ideas and feelings (Ash & Clayton 2004; Clarà, 2015). Nevertheless, if professionals and school faculties improve their own reflective skills by applying them in their own lives, reflective practices can be modelled more effectively to students (Rogers, 2001).

2.2 Consensus decision-making

In general, everyday situations require the process of deciding. This process can be done through two different practices, by either voting or reaching a consensus (Bressen, 2012). Consensus decision-making - also referred to as consensus building (Seeds for Change, 2020) - is a process in which a group works together to reach a shared understanding and agreement on a particular issue. It is a collaborative approach that aims to ensure that everyone's opinions and concerns are heard and considered, and that decisions are made through a process of dialogue and compromise (Bressen, 2012; Seeds for Change, 2020; The Consensus Council, 2018). The goal is to reach a decision that meets the needs and interests of all parties involved. According to Susskind et al. (1999, p. 6),

consensus is “A process of seeking unanimous agreement. It involves a good-faith effort to meet the interests of all stakeholders. A consensus has been reached when everyone agrees they can live with whatever is proposed after every effort has been made to meet the interests of all stakeholding parties”. Because it considers the opinions of each individual or each of the parts equally, consensus is more advantageous in comparison to voting. In consensus decision-making processes resolution is facilitated and the sense of unity increases group bonding. Whereas in voting, a negative atmosphere can be experienced from the sense of having winners and losers (Bressen, 2012).

Consensus decision making is often used in situations where the group members are working towards a common goal, such as in a community or organizational setting (Bressen, 2012). For instance, group consensus is a crucial process in cities’ development plans and management. This process allows participation of multiple players, including citizens and stakeholders in important sustainability-related decisions for the local community, such as economic and community development, and natural resources protection (Innes, 1992). Still according to Innes (1992) for the group consensus to happen some conditions need to be met. It is necessary to indicate the interests that are at stake, as well as call upon stakeholders who are concerned with such interests. In complex decision-making, experts should be invited to communicate with other members of the group so they can have a broader perspective of the issue. However, it must be clear that every participant, regardless of their roles outside of the group, has equal representativity in the decision.

In the context of climate change policy, consensus decision-making is often used in international negotiations, where countries come together to agree on actions to address climate change. The United Nations Framework Convention on Climate Change is one such forum where consensus decision-making is used. For example, the Paris Agreement, which was adopted under the UNFCCC in 2015, was the result of a long consensus decision-making process. Negotiators from over 190 countries worked together to agree on a set of commitments and actions to reduce greenhouse gas emissions and limit the rise of global temperature to 1.5 degrees Celsius (Bernardo et al., 2021). The agreement was reached through a series of negotiations and consultations that lasted 15 years, in which all parties were able to express their concerns and opinions and work on a common goal (Bernardo et al., 2021). More recently, in the Conference of the Parties, in 2022 (COP27), parties struggled to find a common ground regarding the next steps to ensure mitigation of climate change and financial repair of vulnerable countries. After overcoming the high political disputes at stake, they finally achieved a historic consensus, establishing arrangements for a funding for loss and damage for the countries that

are most threatened by climate change (United Nations Framework Convention on Climate Change, 2022).

In the context of education, consensus decision-making is important because it promotes a series of benefits for students and teachers. Consensus encourages students to share their opinions in an environment that promotes respectful and open dialogue, with consideration to different viewpoints (Seeds for Change, 2010). Thus, it can lead to better learning outcomes and a more inclusive classroom environment. As consensus aims at reducing the hierarchy of decision-making, it can promote a sense of shared authority and responsibility amongst students (Sartor & Brown, 2004, Seeds for Change, 2010). In a classroom with consensus everyone has the power to decide or to refuse decisions. According to Sartor and Brown (2004) when students understand their shared role in the decision-making in the classroom, they can monitor themselves towards their pre-established agreements, mindful of the reasons why these agreements were made. The share of authority extracts the teacher from the position of central figure with a depository role (Freire, 1993) allowing for a participatory and meaningful learning experience. Pupils are then not only free but also empowered to use their creativity and work in collaboration to solve problems. Nonetheless, consensus can also help to develop students' critical thinking, and communication skills (Bressen, 2012; Sartor & Brown, 2004; Seeds for Change, 2010).

Considering the aforementioned, teachers can use consensus decision-making in a variety of ways to promote student engagement and collaboration, such as:

A. Classroom rules and expectations: Teachers can use consensus decision-making to establish classroom rules and expectations with their students, as this can help to create a sense of ownership and responsibility among students, as well as promote a positive classroom culture (Sartor & Brown, 2004).

B. Group projects: Teachers can use consensus decision-making to help students work together effectively on group projects. This can involve setting group goals and expectations, determining how work will be divided, and resolving any conflicts that arise.

C. Curriculum design: Teachers can use consensus decision-making to involve students in the design of the curriculum. This can help to ensure that the curriculum meets the needs and interests of the students and encourages active engagement in the learning process (Jagersma, 2010).

D. School governance: Teachers can use consensus decision-making to involve students, parents, and other stakeholders in school governance decisions. This can help to promote a sense of community and shared responsibility and ensure that decisions are informed by a range of perspectives and expertise (Rinaldi, 2016).

Overall, consensus decision-making can be a valuable tool for teachers to promote student engagement, collaboration, and active participation in the learning process. It can help to create a more inclusive and democratic classroom environment, where everyone's voice is heard and respected.

2.3 Climate change & CCE

Climate is the long-term weather conditions of a particular region, which is naturally influenced by many different factors such as geographic location, atmospheric conditions, and seasonality. Although it is a variable phenomenon, its changes are hardly perceived in the short term but rather over long periods of time (Cowie, 2013). Climate and biology are intimately integrated. All forms of life are dependent on climate conditions to thrive, certain species require more specific conditions than others, and minimum disruptions can be enough to affect the survival rate of an entire group. Through a feedback mechanism, living specimens contribute to climate, influencing vital cycles such as the water and carbon cycle. (Climate University, n.d.)

Climate science is dated as early as 1856 when the female scientist Eunice Foote described the influence of certain gases, such as carbon dioxide and water vapor on modulating the solar heat in the atmosphere (Sorenson, 2011). A few years later, more sophisticated experiments conducted by John Tyndall allowed for quantification and description of the heating-absorbing properties of some gases, resulting in the recognition that a variation in concentration of certain gases could lead to changes in climate (Cowie, 2013; Sorenson, 2011). In the mid-20th century, scientists began to observe changes in the Earth's climate that they attributed to human activity. The burning of fossil fuels and deforestation were identified as the primary sources of greenhouse gas emissions, which contribute to global warming (Cowie, 2013).

In the late 1970s, the scientific community began to demonstrate concern about the global impacts of climate change, and the first World Climate Conference was held (Gupta, 2010). In the following years there was an increase in the alarm about climate change, with a consequent increase on research and documents that would address sea levels rise, severe heatwave and storms, and the extinction of many

plant and animal species (Cowie, 2013). In 1988, The United Nations established the Intergovernmental Panel on Climate Change (IPCC) which was the first official report to provide scientific consensus on climate change, concentration of greenhouse gases and suggested potential mitigation for the impacts (Gupta, 2010). Since then, there have been numerous international agreements and initiatives aimed at reducing greenhouse gas emissions and mitigating the impacts of climate change. These include the United Nations Framework Convention on Climate Change in 1992, the Kyoto Protocol in 1997, and the Paris Agreement in 2015 (Cowie, 2013; Gupta, 2010).

Climate mitigation and adaptation are two current approaches to address the challenge of climate change. Climate mitigation refers to human actions taken to reduce or prevent greenhouse gas emissions, aiming at limiting the magnitude of climate change (Climate University, n.d.). This can be achieved through a variety of measures such as a transition of the current energy systems to systems using only renewable energy sources, promoting sustainable transportation, reducing deforestation and forest degradation, and enhancing agricultural practices (IPCC, 2014). The IPCC has highlighted the importance of reducing greenhouse gas emissions to limit the impacts of climate change stating that "emissions of greenhouse gases from human activities are the largest driver of observed climate change since the mid-20th century" (IPCC, 2014). Currently, most of the emissions are attributed to countries in the global north, due to their high demands in energy systems, land use and cover change (LUCC) and transportation. (Climate University, n.d.) The conflict of interests associated with sovereignty of countries make negotiations for mitigation a challenge, even though there is evidence of the importance of mitigation in the improvement of people's livelihood, through the direct impact in their health and wellbeing as well as in economic activities (IPCC, 2014).

Climate adaptation, on the other hand, aims to reduce the vulnerability of societies and ecosystems to the impacts of climate change that are already happening or are expected to occur in the future. As changes in climate are inevitable despite mitigation actions, adaptation emerges as a key factor to anticipate future changes (Climate University, n.d.). The IPCC (2014) has emphasized that while mitigation is crucial to limit the magnitude of climate change, adaptation is also necessary because some impacts of climate change are already unavoidable. Füssel (2007) defines that adaptation can be proactive when actions are done in advance, or reactive, after the damage has happened. Adaptation can involve a range of measures that could be technical such as building barriers to protect against sea level rise or developing drought-resistant crops, but also educational, since adaption actions are embedded in knowledge and skills (Füssel, 2007; IPCC, 2014).

Climate change education (CCE) is the process of equipping individuals with the knowledge, skills, values, and attitudes necessary to understand and act toward the complex issues related to climate change (UNESCO, 2022). It can take many forms, including formal education in schools and universities, community outreach programs, and public awareness campaigns. It can cover a wide range of topics, including the science of climate change, the impacts of climate change on ecosystems and human societies, and the policies and actions needed to address climate change. CCE should also emphasize the importance of sustainability. (UNESCO, 2022)

Education is seen as an essential factor to understand and tackle climate change (UNESCO, 2019). According to Ouariachi and Elving (2020) education, especially climate change education, is aiming to increase knowledge and develop decision-making and awareness, but also encourages people to make changes in attitudes and behavior to start making sustainable choices. To get individuals to consider their choices and make lifestyle changes, more effective strategies are needed to increase reflection and discussion around climate change.

As a sub-area of Education for Sustainable Development (ESD), CCE should promote a holistic perspective of such complex matters, comprising the multiple dimensions harnessed to climate change and sustainable development, such as the environmental, economic, social and cultural aspects (Osman et al., 2022). According to UNESCO (2020) ESD is an intrinsic part of the Sustainable Development Goals agenda, particularly addressed in target 4.7.

> *"By 2030, ensure that all learners acquire the knowledge and skills needed to promote sustainable development, including, among others, through education for sustainable development and sustainable lifestyles, human rights, gender equality, promotion of a culture of peace and non-violence, global citizenship and appreciation of cultural diversity and of culture's contribution to sustainable development."* (UN, 2015, p.17).

Experts alert that the next generation of students need to be prepared to deal with the tough reality caused by the increase in global population, food scarcity, loss of biodiversity and other catastrophic consequences of the climate crisis (Mochizuki & Bryan, 2015). Although for some, climate change is still a fallacy, climate change education emerges as a tool against political manipulation, media and fake news (Boon, 2016). Climate change education is necessary to not only promote understanding about the direct impacts of climate linked to nature but also to recognize its systematic link to structures of power and a capitalist system (Mochizuki & Bryan, 2015, Stevenson et al., 2017). By acknowledging the

materiality of climate change, one can then analyze their own attitudes improving their relationship with nature. In that sense, climate change education can and should provoke critical thinking, enabling people to become responsible citizens who have agencies to determine appropriate responses to climate change (Boon 2016, Dillon et al., 2016). These critical skills support students' reflection and decision-making on acting more ethically with the world they live in (Mochizuki & Bryan, 2015). Furthermore, because of the growing necessity for ESD, sustainability related courses are becoming part of the curriculum in higher education around the world. (García et al., 2006)

2.4 Gamification

The use of technology has increased significantly in all areas during the 21st century, which has paved the way for gamification. Modern learners are used to using the internet to watch videos and play games with their computer or smart phones (Mawhirter & Garofalo, 2016). It is almost inevitable for individuals to encounter play in our modern societies. The popularity of games lies on their intellectual challenge and entertainment (Piñero Charlo et al., 2022). Thus, gamification has become a trend in many sectors in recent years (Caponetto et al., 2014). Gamification is characterized as a strategy to bring gaming and playful elements into different environments to increase engagement and motivation to participate in activities and actions. By achieving this, gamification relies on utilizing badges, points and leaderboards. Furthermore, implementing gamification in an environment can reduce stress and increase unity among people. Although, it can be challenging to apply gamification in practice (Vesa, 2021). Gamification has received more attention in the past years but is still a fairly young topic to study. Therefore, there is still much to learn from the method and its benefits (Hamari et al., 2014).

Games are co-produced by the game developer(s) and player(s). Game developers oversee the storyline creation and create rules and game patterns, whereas the players play and interact with the game itself. It is essential that the gamified activity offers suspenseful and challenging experiences for the players to strive for successful experiences (Huotari & Hamari, 2012). Hamari et al. (2014) found in their study that gamification has been mostly implemented in educational contexts and its effects on learning outcomes has been perceived mainly positive, especially with motivation. Although, it is noteworthy that gamifying activities can increase competition, thus producing some negative outcomes. However, they conclude that gamification generally works.

2.4.1 Gamification in education

School environments can often result in undesirable outcomes such as disengagement and dropping out, since school activities are often dull and lack playful experiences (Lee & Hammer, 2011). Therefore, gamification has increased in educational environments since it can enhance students' engagement similarly like games (Smiderle et al., 2020). Educational games support collaboration and learning (Mawhirter & Garofalo, 2016). According to Strickland and Kaylor (2016), experiential education facilitates skill development and learning when being involved in an active learning environment and given the time to reflect. Based on their research, gamified learner-centered activities provide an opportunity for students to be responsible for their own learning. Moreover, game-based learning increases positive feelings, teamwork, creativity, and critical thinking and improves academic results. Building on that, Siegle (2015) described that educational games are often enjoyable and successful when the "stakes are not too high, teams are recognized, all students have an opportunity to win, there are no consequences related to scores, and games target essential academic content" (as cited in Mawhirter & Garofalo, 2016, p. 133). Gamification can offer opportunities to develop professional skills and learn meaningful content, since it is known that individuals learn better when they enjoy what they are doing (Fabricatore & Lopez, 2012).

Lee and Hammer (2011) describe that gamification offers other benefits as well than increased motivation and engagement. Games develop and enhance students' emotional, cognitive, and social capacity. They elaborate that by encountering failure and seeing it as an opportunity, students can eventually reach success and assess their own abilities through games. Games can guide students to explore new techniques to learn and challenge their thinking. Moreover, games can challenge students to try new roles, enhance interaction and develop 21st century skills. Gamified learning environments can also be beneficial in supporting knowledge and skill acquiring through fun. It allows individuals to deal with ill-structured problems and the development of events. Furthermore, gamified learning increases the use of various thinking processes (Fabricatore & Lopez, 2012).

Even though gamification can have many benefits, it also has some challenges that should be considered. According to Lee and Hammer (2011) utilizing gamification can steal teacher resources, thus increasing the workload. It can also increase student expectations and get them to think that learning should involve some sort of reward, which can generate laziness. Moreover, they mention that gamifying learning content has the risk of taking playfulness away from the experience and destroying

the idea of games. Additionally, being involved in an active learning situation as a group can affect group dynamics negatively, especially students' participation (Strickland & Kaylor, 2016).

2.4.2 Educational escape rooms

According to Nicholson (2015, p.1) escape rooms are "live-action team-based games where players discover clues, solve puzzles, and accomplish tasks in one or more rooms in order to accomplish a specific goal (usually escaping from the room) in a limited amount of time". Escape rooms require teamworking skills and critical thinking. The gamified learning environment provides a fun experience to interact and learn with other individuals and it is suggested that team members should have various personalities and thinking skills, to make the experience more successful (Adams et al., 2018). One of the biggest learning outcomes from escape rooms are teamworking and communication skills. Successful teams can communicate well to tackle the challenges during the experience (Nicholson, 2015). The benefit of escape rooms lies in their flexibility and the ability to affect multiple learning skills. Escape rooms can be integrated into any curricular content through instructions, puzzles, problem solving and challenges (Moura & Santos, 2020).

Escape rooms can be beneficial for learners who prefer a kinesthetic learning approach where they can complete hands-on activities. Moreover, visual learners can benefit from the visual cues and resources in an escape room (Adams et al., 2018). If the game uses a story or a theme that has local interest, players have a chance to learn more about it (Nicholson, 2015). The variety of props and use of real-life occurrences can make the players feel like they are part of the narrative, thus getting them more engaged and increasing internal motivation to manage the game (Mac Gregor, 2018). Furthermore, when the storyline and the content of the game is carefully considered and there is consistency between tasks, the experience is most likely more compelling to the players (Clarke et al., 2017).

Special attention should be given to the beginning of the escape game. Players should be given enough guidance in the beginning for them to be able to know how to proceed and how to reach the goal (Lopez-Pernas et al., 2019). Nicholson (2015) describes that escape rooms usually include a gamemaster who supports the players by offering them hints if needed and making sure that the experience is fair for each participant. He adds that it is essential for the gamemaster to offer enough time for the players to explore, learn and engage in problems before presenting them with the hint. Furthermore, since the experience can raise many emotions, it is important for the gamemaster to

arrange time for debriefing for the players to discuss their feelings and thoughts about the game. In their study, Veldkamp et al. (2021) found that teachers perceive reflection to be a boundary condition for learning and consider that enough time for reflection is needed after an experience like educational escape room.

Escape rooms have become an interest in science-based teaching and learning environments, since students' interest in studying science has decreased. This has negatively affected students' abilities and knowledge base, such as critical thinking and problem-solving skills, which are important 21st century skills. Escape rooms are believed to enhance these 21st century skills (Li & Tsai, 2013). According to the study of Veldkamp et al. (2021) escape rooms were perceived as a positive method to learn about science but most importantly to develop collaboration and teambuilding. Students were able to stay engaged and interactive, and they also reported learning new knowledge. Moreover, most teachers perceived escape rooms as an appealing learning environment and would use the method again. They mentioned escape rooms as a good tool to specifically increase motivation, but most did not believe it to be suitable to build knowledge. To build knowledge in science, they believe it requires a more structured environment and even more reflection to learn. Another challenge found was that it can be difficult for teachers to find time to develop the game.

Nevertheless, according to Ouariachi and Elving (2020) educators should provide strategies where individuals are encouraged to reflect to understand various issues. They suggest that escape rooms can offer an interactive solution for individuals to discuss wicked problems such as climate change. Escape room provides a setting to engage, communicate and think critically to find solutions to challenges. Involving a timer, which most escape rooms do, can raise the sense of urgency and motivation to act and understand how important it is to start acting more sustainably.

3. METHODOLOGY

This chapter will introduce the methodological approaches used in the research and describe the research ethicality and reliability. Next, we will present our research questions and methods. This includes the data collection procedures, target group and the analysis process.

3.1 Research questions

The purpose of this research was to assess pre-service teachers' reflection and consensus decision-making by participating in a climate change themed escape room. The research was aiming to answer the following research questions:

- Is gamification within an escape room setup functioning as an educative tool to facilitate discussion about climate change among pre-service teachers?
- How is the consensus decision-making done about climate change among pre-service teachers?
- How is the reflection process done in groups, and to what extent can pre-service teachers reflect on climate change within an educational escape room?

3.2 Qualitative research

There are different methods to be used when conducting research and collecting data. For this study, the most appropriate way was to use qualitative methods. Qualitative research can involve opinions, beliefs, and feelings, which become essential when trying to understand human behavior and how people act (Greetham, 2014). These points fit with our research goals. The research was examining how university students reflect, share their opinions and work together to create consensus.

According to Taylor, Bogdan and DeVault (2016) qualitative research is defined as descriptive data, which describes observations of people, target phenomena and behavior. They add that qualitative researchers often spend time in different settings to build understanding of the phenomenon and different patterns in the data to create research conclusions and explanations. Thus, the researcher is in contact with specific situations to be able to perform data collection such as observations, analyses, and reflections. This allows the researcher to gain a deeper understanding of the phenomena and contexts. In addition, qualitative research data is often in the form of words, not numbers. King, Keohane and Verba (1994) point out qualitative research to focus more on one or a small number of

cases. In this case, the research only involved a total of 23 participants. While qualitative data can provide large amounts of data, it can also create stronger and in-depth research and reduce mistakes in the study. Additionally, qualitative studies usually focus on a specific issue, event, decision, institution, or location.

The first phase of qualitative research procedures is to create research questions from what is studied. The second phase includes data collection, such as observations and interviews, which were essential parts for this research. The third phase includes making decisions on a sample, selecting participants from a population, also determining a setting. The fourth phase consists of data analysis, which usually lasts through the whole investigation. Making conclusions and understandings from the data creates the final phase, where the researcher is making interpretations and evaluations from the study (Martella et al., 2013). Each phase was carefully considered throughout this study.

3.3 Game development

The escape room game was initially designed by the authors of this study as a project for the course Education for Sustainable Future as part of the master's program in Pedagogy and Teaching for Sustainability at the University of Eastern Finland. This experimental project was considered a pilot study in which the researchers tested and evaluated the tasks and observed outcomes. After running the experimental project, adaptations were made to fit new research questions.

The game preparation was conducted in three main phases: the research phase, the design phase, and the implementation phase. The research phase was constituted of literature review about gamification and escape room pedagogy, as well as product research of the current escape room games and models available in the market. Features such as content, knowledge level, playability, time, and feasibility were taken into consideration. After recognizing the essential features of escape rooms, the designing phase was initiated. In this phase, four tasks were created from the ground up, meaning that apart from scientific content, everything was created by the authors of this study. The tasks will be detailed further. The implementation phase consisted of organizing a schedule for the participants, setting up the scenarios in the rooms and handmake the flashcards used in the tasks. All of the materials were provided by the SM4RT LOC laboratory.

The game was designed to last one hour per group, in which four tasks should be completed. For all of the tasks, the participants would find an envelope containing instructions for the task itself and the

flashcards to use in it, and another envelope containing a riddle to be solved. The riddles were encrypted messages or puzzles that would lead the participants to the next task. The storyline that conducted the game was based on the United Nations climate conferences. In the game, the participants were simulating the United Nations' representatives who gathered to discuss and develop solutions for climate change and climate change education. A more detailed description of the tasks can be found in appendix 2.

Task 1. "The investment"

In this task, the participants received seven flashcards with main topic areas for mitigating climate change, including: education, energy security and renewable energies, affordable and sustainable transportation and mobility, improved air quality and health outcomes, adaptation measures to climate disaster, sustainable food and agricultural systems, gender equality actions. They also received three coins that symbolized investment money. The main task was to debate and come to a consensus about which areas would receive their investment, clarifying the reasons for the groups' final decisions.

Task 2. "Fake or Fact"

The participants received printed copies of two online-news articles, including the authors, the name of the website and date in which they were published. Participants needed to decide if and which of the articles were fake or fact, also stating the reasons that led them to draw conclusions.

Task 3. "Curriculum planning"

In this task the participants received quotes from the Finnish National Core Curriculum (Finnish National Board of Education, 2016) regarding sustainability and climate change in basic education. They should discuss and decide which strategies they could use to implement climate change education in primary school lessons.

Task 4. "Evaluating values"

The participants received nine flashcards. Seven of them contain factors that influence climate change action, according to the research of Cantell et al. (2019). The participants were expected to rank the factors, in order of relevance, deciding as a group which of these factors were more important or influential in climate action. The ranking structure was open to interpretation of the groups. The two other cards were meant to be used as "final decision" cards, in which they had to decide if climate change action is a matter of individual or collective actions.

3.4 Data collection

The game took place in the grounds of University of Eastern Finland. The university has an escape room laboratory, named “SM4RT LOC”. The laboratory is located in rooms E001 and E002 in Educa building on Joensuu campus. The facilities were designed to create a learning observational space for educators and academics. One of the rooms works as the control center, where the game can be monitored through screens. Sound systems and microphones allow communication with the participants at any stage of the game. There is a light system that controls the lights of the other room. The lights can be adjusted according to the atmosphere of each stage of the game. The room also has equipment to measure carbon dioxide levels, however, this component was not assessed in this study. The second room is where the experiment takes place. It is divided into three smaller rooms, which are locked through digital and regular locks. Each of the smaller rooms is equipped with television screens in which important information and the time can be exhibited.

The data of this study was collected through observations, interviews, and video recordings. All the groups were recorded during the entire duration of the game, totalizing five hours of video recordings. The interviews were conducted in the same facilities, immediately after the end of the game, with each group. Participation in the game was voluntary and every participant signed a consent form.

3.4.1 Target group

An essential part of data collection is to select the target group for the research, who to observe and interview (Taylor et al., 2016). Reflection and decision-making skills are necessary skills in teaching occupation and teachers need to obtain quality teamworking skills to be able to work with other teachers, parents, and other school members. Thus, this study targeted higher education students, particularly Finnish students who are studying to become future teachers, so called “pre-service teachers”. There were no specific requirements regarding the main subject of their studies. However, all the students were enrolled in the course Education for Sustainable Future. The students were assigned into groups by the course teacher who informed them about the chance of participating in escape room research. Five groups of four to five people participated in this research, totalizing 23 participants. Despite the participants' first language being Finnish, the experiment was entirely

conducted in English. The participants were pre-informed about the language requirements before signing up for this study.

3.4.2 Observations

Martella et al. (2013) describe observations as descriptive data from a phenomenon or a situation, where the researcher has observed participants in a certain setting. Observations allow the researcher to understand a situation, analyze it and to draw conclusions about it. Although, it is important to remember that observations are usually used in conjunction with other methods to extend the research analysis (Blaxter et al., 2010). Field notes are an essential part of observations, which are notes from what is seen and heard in research situations. Field notes need to describe the physical setting, describe what is going on and include participant feelings and actions. Field notes can be in the form of written notes, audiotapes, videotapes, or photographs (Taylor et al., 2016).

All target groups were observed during the game through a camera and notes were taken from the research situation and participant actions. Additionally, the events and actions were recorded to a videotape including audio, which were later transcribed, coded, and categorized into themes. By closely observing the research situation, it allowed the researchers to offer help and guidance when it was needed. When the participants showed distraction or were not able to proceed, the researchers gave extra hints through a speaker for the groups to realize the next step.

3.4.3 Interviews

Interviews are an essential part of qualitative research methods. Although, it should be remembered that interviews do not only belong to qualitative research methods but also to quantitative (Eskola & Suoranta, 1998). According to Rapley (2004) the idea of the interview is to receive in-depth descriptions about matters, which the interviewer can use to create assumptions and detailed results. Therefore, the interviewer is researching specific topic in which s/he wants quality data to make research conclusions. Interview is one alternative and usually enhancing method to gather information for research purposes. They are also a good addition to observations since the researcher might not be able to observe every single matter. Thus, interviews offer answers and clarifications, which are essential in creating research results (Martella et al., 2013).

The researchers regarded interviews to be an essential method to gain clarifying answers about the group's reflections and opinions about the game topic. Therefore, interviews for the research were conducted as semi-structured group interviews with four individual interviews as an addition to supplement participant answers. Group interviews can be beneficial since participants can influence each other positively by encouraging everyone to get engaged in the discussion and reflect more. Additionally, if one has challenges in reflecting the other group members can offer help (Eskola & Suoranta, 1998). These were specifically considered in the interview planning process, since the interviews were conducted in English, which was the second language of each participant. Our goal was to create a comfortable space where everyone was heard, treated equally, and provided enough time to reflect. Participants and researchers offered help to each other if it was needed.

3.4.4 Video recordings

It is thought video and photographic material to be part of qualitative methods, which provide powerful data of real-life events (Loizos, 2000). Taylor et al. (2016) describe that it could be challenging for a researcher to make valid conclusions about phenomena, actions and people from one-time meetings or observations. Thus, video recordings become beneficial since they can support observations and provide data that could be left unnoticed. Video recordings allow researchers the chance to produce deeper interpretations about events and people or notice patterns that words cannot. Combining video recordings with other qualitative research methods can provide abundant data, which can strengthen the understanding of events and data. Since the video recordings were used to substantiate observation and interview notes, it can reduce data errors and create more valid and reliable research results. The recordings allowed the researchers to go back to the recording material and correct misconceptions and to find other meaningful insights that were essential for the data evaluation.

3.5 Ethical issues

The first important part before starting the data collection is to inform the participants about the research and come to an agreement with them on taking part in the research. Participants must be aware of the research procedures, management and understand what the research is about. (Blaxter et al., 2010). Before the data collection started, all participants were handed a paper consent form, which

included all information about the research and how the data will be used. Participants were given the time to go through the form and in the end if they agreed with the terms, they were asked to sign the consent form. Each participant agreed with the terms and gave their consent to be part of the research. At the start of the game, the researchers still went through the game procedures, what the research was about and made sure that everyone was aware of what to be expected. It was also made sure that participants were able to quit the game and leave at any time, if they felt so, which ensures the research voluntary.

Protecting participant identities was also considered during the whole research process, which is one of the main ethical concerns (Blaxter et al., 2010). All data is anonymized, and participants' identities are protected at all times. Recordings and all other data are only available for the researchers of this study. Recordings will be used in the research to present results but are not visibly published or used in any presentation. To further protect participant identities, all research data will be destroyed after completing the study. Additionally, consent forms are stored and protected by the researchers, and later destroyed accordingly. The study's research ethics were informed to and approved by our university research supervisor to ensure the rightful research procedures and actions.

3.6 Research validity and reliability

To ensure the research validity and reliability the researchers have utilized multiple methods to gather and analyze data, which will lead to more valid and reliable interpretations and results (Golafshani, 2003). It is commonly said that qualitative research often lacks objectivity and can involve researcher bias, which then affects research validity. Triangulation, the use of various methods, can be helpful to minimize such issues and errors in the data. Triangular techniques are often beneficial when trying to organize and explain human behavior. (Cohen et al., 2007). This research utilized observations, video and audio recordings and interviews in data collection to validify and increase the reliability of the results. Moreover, to have two researchers collecting and analyzing research data can increase the validity of the research, since it can decrease data selectiveness. Additionally, if both come to same conclusions and there is no conflict between the material, it increases the data reliability (Eskola & Suoranta, 1998). To exemplify, the researchers worked individually during the writing process but communicated continuously when reporting the research results to make sure both arrived in the same

conclusions. Problems, obscurities, or different opinions were solved together by talking them out, pondering solutions or asking for advice from supervisors.

To reach greater validity in research, interviews should also be carefully planned and structured to minimize bias (Cohen et al., 2007). This was considered when planning the interview questions. Even though the interviews were semi-structured and allowed the researchers to ask clarifying questions if needed, all groups still answered the same set of questions (appendix 1). The wording in the questions was carefully thought-out, in which it was not leading interviewees in any certain direction to seek preconceived answers.

3.7 Data analysis

To make sense of the research, a model was used to analyze data and to assess reflection. Koole et al. (2011) point out that there is vagueness on how to best assess reflection and there are different models that represent the reflection process, which makes the assessment challenging. On the other hand, Koole et al. (2011) presented a comprehensive model that combines what the other models have in common. In this so-called "eclectic model" they describe reflection as a cyclical three-step process (figure 1). Each of the steps are analyzed through two indicators of reflection. The process begins with the experience that will trigger one's reflection. Then, the first step is to review the initial experience, considering one's ability to properly describe an event or situation, and to identify essential elements of the original experience, being able to describe one's own thoughts and feelings. Secondly, it is the critical analysis step, in which one's ability to both ask and answer searching questions is assessed utilizing frames of references. Lastly, it is the moment for the reflective outcomes, evaluating one's ability to draw conclusions and describe concrete learning goals and plans for future action. Therefore, the researchers chose to use this model in the research to assess and analyze participants' reflections. The model was used to analyze participants' reflections during the game and when they reflected on the game experience at the interview. Interview questions were largely formed based on the assessment model.

The analysis process began by carefully transcribing data from the recordings and becoming acquainted with the data by reading it several times. Afterwards, we reported and analyzed the results by utilizing the reflection model by Koole et al. (2011). When reporting results and quoting participant

answers, we used abbreviations for group=G and person=P, also numbers to state which person in that group had reflected.

Table 2 Operational indicators of the reflection process

Aspect of the reflection process	Indicators	
Reviewing the experience	1.	The ability to describe an event/situation adequately.
	2.	The ability to identify essential elements and to describe own thoughts and feelings
Critical analysis	3.	The ability to ask searching questions.
	4.	The ability to answer searching questions and being aware of the frames of references in use.
Reflective outcome	5.	The ability to draw conclusions.
	6.	The ability to describe concrete learning goals and plans for future action.

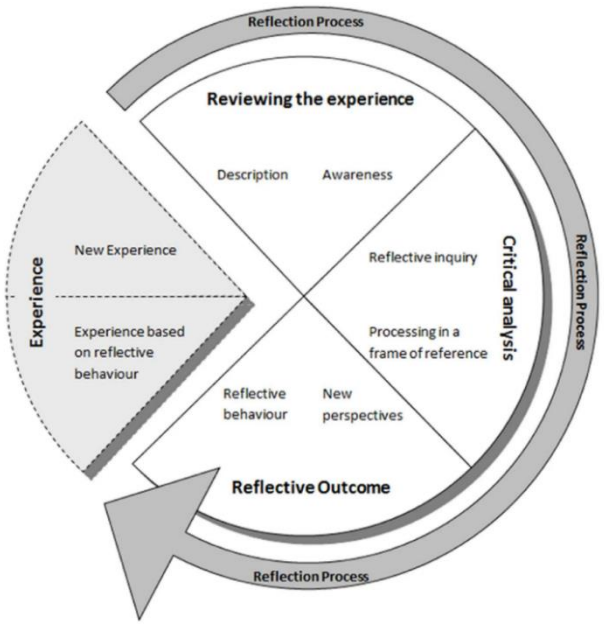


Figure 1. A descriptive image of the reflection model and the indicators of each stage of reflection process. Adapted from Koole et al. (2011).

For consensus decision-making, data was analyzed through an inductive approach. No model was used to evaluate the process. Instead, there was a search for indicators that displayed the groups' agreement. Discussions, participation, words, and gestures were analyzed during the game execution, and if any outstanding situation regarding the consensus process occurred, it was brought up during the interview after the game. The data was reanalyzed through video recordings, transcription of the recordings and interviews.

4. RESULTS

In the results, we aimed at identifying and reporting the different levels of reflection on the topics, considering the levels proposed on the Koole model. In addition, we outlined the different forms specific participants expressed their reflections, whether they demonstrated to reach partial stages or complete reflection. In that sense, we considered participants' ability to describe the events properly, including a description of essential elements, being critical with their justifications and drawing conclusions.

During the experiment, participants acted and behaved in different ways. Some demonstrated themselves to be more open and participative, engaging in full discussions, whereas others adopted a more passive posture. Likewise, there were different profiles of players in the game. There were players who were focused on completing the tasks carefully, centralizing the discussion as a fundamental part of the game, and others who focused on the playability, aiming at completing the tasks and moving forward according to the clock. As the game developed, some roles were intuitively designated to certain participants. For instance, most groups had a leader who took the initiative to read the instructions - and support the understanding of certain terms. Depending on the tasks this role switched, and people restructured the group dynamics.

In the beginning of the game, it was clear that most of the groups experienced some nervousness and awkwardness. There was a lot of nervous laughter, ignorance and even silence. Participant G3P4 even expressed that she was not comfortable in participating and sharing her own thoughts because one person was a stranger. But as the game continued, the participants were able to act more comfortably.

Overall, few participants presented complete reflection during the game. In groups 1, 2 and 3 at least one participant took the lead in terms of initiating discussions, pondering, and reflecting. The other participants from these groups were still able to contribute to the tasks to some extent, fluctuating between inconclusive thinking and intermediate reflection. In two groups, groups 4 and 5, the discussions seemed to be more evenly distributed among the participants, with occasional alternance of leadership according to the topic in debate. Group 4, however, demonstrated more consistency in this regard. All groups used code switching and translations occasionally to make sense of the topics and what the tasks required them to do, to make clarifications and to help other participants to get involved.

As a matter of organization, the data results for reflection will be presented task by task. We decided to do so since the reflection patterns and their quality differed depending on the task. This was also a clearer way of presenting data.

4.1 Game reflection

Task 1

The goal in the first task was to encourage the participants to choose areas they valued and saw as most significant to climate change mitigation. All groups were unanimous about choosing education as one of the most valuable factors contributing to climate change mitigation. In all groups participants attributed this choice to the fact that they are all pre-service teachers, therefore seeing the importance of educating people for future action. Energy security and renewable energy systems were also one of the most valuable factors according to the three groups. All groups acknowledged all the fields proposed as valuable areas to be invested in. Participants G1P1, G2P1, G3P1, G4P3, G5P3 clearly expressed that the fact they are all teacher students, or that educated people understand what needs to be done, which justified the choice of education.

However, groups 2 and 3 demonstrated to have difficulties in presenting arguments to support their choices and opinions, recurring to obvious answers. At times, it was explicit that some participants from groups 2 and 3 lacked knowledge on what should be discussed when stating such as "*I really don't know much about these things*". However, it was not always clear if they were expressing to have a language barrier or simply could not provide arguments due to lack of specific content knowledge. This affected the groups 2 and 3 reflection quality, and they were not able to obtain all levels in their reflection, since they lacked justifications and content.

In groups 1 and 5, the participants who most actively engaged in the discussions also demonstrated logical reasoning on the game's aspects. In both groups, participants utilized logic as a tool to decipher the codes that would lead to the next levels of the game. Although the clues were intendedly easy to find, few people could see what sometimes was "literally" in front of them. Participants G1P1 and G5P3, on the other hand, easily understood the simplicity of the game and used it favorably to their own groups. For instance, G1P1 and G5P3 associated the text color of the riddle card with the numbers, also in red, in the instructions for task number 1, which rapidly led their groups to the way out to the second task. Also in task 1, G5P3 used logic to understand symbols, realizing that although the coins provided in the task differed from each other, they represented equal value in their decision.

Similarly, both G1P1 and G5P3 demonstrated to have a broad world view, showing knowledge of current global issues of sustainable development, and understanding how certain matters are interconnected. In

task 1, G1P1 pointed out how gender issues are intensified in certain countries, addressing economic aspects to it, and a special concern to women rights. She was able to assess her framework of reference and personal experience as a woman to draw conclusions:

"Gender equality actions. Personally, I would pick it cause I'm a woman and I would love it if there was gender equality […] gender equality also comes like with poor people, rich people, everything, just equality altogether...just gender equality, especially how women are treated in some countries."

G5P3 addressed the necessity to understand the current climate issues, to have people equipped to work on mitigation of climate change. This participant supported her opinion with critical thinking and provided concrete examples, demonstrating the use of scientific background knowledge in her reflection.

"It's important for people to understand what the world is going through and understand how we can mitigate climate change […] less CO2 will mitigate climate change. Also, sustainable food and agricultural systems, because there's so many people on this planet that we need to be able to like, feed people, and it has to be in a sustainable way, it cannot just, you know cut down all the forests, to have land to have food."

Additionally, to groups 1 and 5, group 4 developed conversations that demonstrated to have some reasoning and argumentation. They were able to make connections between the areas of value, which helped them justify their investment choices. Despite showing certain coherence, participants left vague statements, lacking critical reasoning, for example a more concrete cause-consequence effect, and a reflective outcome with possible solutions for the issues in question. For those reasons, these three groups presented partial level reflections. Participants demonstrated some background knowledge and used it to guide their reasoning and choices. For example, G4P1 stated:

"Yeah, and we can do this (transportation) after this (energy security) so they are like related. We can improve health care as we go but we can't do that if..."

Task 2

In this task, the goal was to identify fake news and debate about them. All participants demonstrated to be well equipped against fake news. Most of them were able to easily recognize elements that are common in this type of news, such as appealing vocabulary and sensationalist headlines. When finding means to justify the searching question, participants mentioned that the writing style of one of the articles

was too aggressive with a strong choice of words, which made them deduce this would be fake. Other participants pointed out essential elements to analyze in an article that would make it trustworthy. Finding evidence from numbers and observing sources and their reliability was some of these elements. Unquestionably, most of the participants relied on their scientific knowledge of climate change to draw their conclusions. As one of the articles contained sentences about climate change being a myth, the obviousness of this statement made it clear for the groups that the article was fake, as G5P2 exemplified in the interview:

"Well, the fake news said that climate change isn't real or happening, so I think for us it was obvious that it's not true. But I guess it can differ from people that maybe some people aren't very understanding and think that climate change isn't happening. And for some people it could be difficult to understand that this is actually fake news if they do believe the news itself"

This reduced the opportunity for the groups to find a more critical stance and question their thinking. Only one of the participants transcended the discussion beyond the facts. Through an ontological perspective, G3P1 questioned the possibility of both articles being real, given their materiality. *"Okay, so, is there anything that is actually fake? Because if this exists, is it fake? Even if it's just any kind of nonsense, it could be just someone's opinion, that doesn't make it fake"*. This turned the reflection into a new direction by considering other aspects and generating thoughtful thinking. However, the discussion did not evolve.

Task 3

In task 3 participants could utilize their knowledge of teaching to think of appropriate strategies to integrate climate change education and education for sustainable development (ESD). An evident strategy among all the groups was to address climate change education to science related classes, for example, biology and geography. Nevertheless, in the matter of sustainability, participants G1P1, G2P2 and G2P3 went beyond and proposed that this topic could be approached in non-science classes, such as home economics and crafts. However, the participants did not provide a detailed explanation of how climate change could be integrated into these classes, thus, reduced the quality of reflections.

Groups 1, 2, 4 and 5 showed similarities in their discussions by raising the importance of combining multiple school subjects together in climate change and sustainability education. They were able to

present quality and various levels of reflection with their discussions. Participant G4P1 reminded that sustainability is "*a multidimensional topic*", considering its social and cultural aspects, which was also addressed by group 2. Participant G1P3 pointed out that climate change can also create "*socioeconomical changes*", alerting that, although climate change might be highly associated with environmental sciences, its consequences can affect different areas. Participant G1P1 and G1P3 pinpointed migratory flows as a consequence of sea level rise and suggested that pupils should be aware of the possible influx of migrants in Finland. They demonstrated the ability to critically think about the interconnectedness of climate change impacts and also describe a concrete future action. Continuing with these thoughts, participant G5P3 reflected about providing pupils with real life examples of how action can be taken inside pupils' own context. This participant emphasized that understanding the global impacts of climate change is important, but it is difficult to measure the reality of other countries. Thus, local actions should be promoted.

"well I think the fact that we are giving real life examples and solutions is way more effective than just say overall in the world because sometimes it is really difficult to grasp or understand when it is said like: "in Africa this is happening" but if you've never been there is really difficult for us, Finnish people to understand what is like to be there in those conditions. So, giving real life examples that we can do here, in our own daily lives is more effective than stating overall."

Such reflection demonstrated critical analysis on understanding that climate change is global but has a diversity of issues that are endemic to each place, which not everyone can relate. It also provided examples of effective future action on teaching practices to meaningfully address the topic.

Despite the heterogeneity of the participants' subject backgrounds, one topic was unanimous among the groups, recycling. There was an inherent association between ESD and recycling, and the sense of action that such topic can raise in pupils. Group 1 suggested that recycling activities are good strategies, because they have performed these activities when they were primary schoolers. In other words, they assessed similar experiences in their framework of reference. Participant G1P1 touched superficially on the possibility of bringing sustainability and recycling to people's own reality. This participant proposed to involve families in learning about sustainability through activities such as an interview. Family participation was seen as significant within participants from group 4 as well. Participants in group 4 debated about the influence of both school and home on pupils' way of thinking. G4P1 and G4P4 were

able to provide arguments and reasoning for their perspectives, and reflected properly while assessing their own experiences:

"*We need to remember that if we are teaching small kids, their parents are usually the ones who impact the most for them, so if the parents are doing differently than we are saying, all the time the mom and dad who has the power, so maybe the teacher's words are not important if they are doing differently at home.*"

"*I'm the kind of person that I learn something at school and home and I'm like: mother we are doing everything wrong we should be doing it this way and she's like, ohh so you say, let's debate and let's discuss and then we find compromises*"

As seen in the first quotation, age was also seen as significant when considering how to integrate climate change education, particularly within groups 4 and 5. Participants G5P1 and G5P3 highlighted that with younger pupils, teachers should be careful about the type of information they provide to their pupils. "*[…] Obviously you don't want to bombard year 3 students with "the world is dying" […] they are gonna get very anxious about it."*. It was proposed that teachers should concentrate more on considering solutions to climate change and to have emotional assessment while educating about climate change to avoid these overwhelming feelings. Moreover, participant G4P5 pointed out that with younger pupils it is important to concentrate on practicality to teach sustainability, for pupils to clearly perceive how their own actions have an effect. Another practical proposal from group 5 was to discuss the importance of knowing how to be critical against climate change related news, which was grasped from the previous task.

Some participants have also debated about implementing strategies that enhance students' motivation. They suggested that motivation could come through activities that do not focus only on the theoretical part of climate change or sustainability, but rather on more engaging activities, as participant G2P3 highlighted *"Something that's not just theory, where they can do something themselves […] something that's fun for them to do"*. Some of the suggestions arose from participants' personal experiences, either as a student or as a teacher. They proposed activities such as games and theme days, gardening, and exploratory outdoor education which could include *"a nature diary, you had to go and find like trees and take the leaves."* and *"Also, how much you can find from nature like berries, mushrooms,*

etc.”. With these reflection points, participants were able to show reasoning and logical thinking, presenting proper and quality level of reflections.

Task 4

All the groups were mainly unanimous that knowledge and thinking skills matter the most when battling against climate change. Groups mentioned that knowledge is the key to acting and to understanding how to act. Showing their ability to answer searching questions and make plans for the future. Participants from all groups demonstrated knowledge about factors that influence climate change actions and were able to find a common ground quite easily. Groups, especially 1 and 4, saw certain factors leading to others, which influenced their decision making. However, reflections from groups 1, 2, 3 and 5 were lacking depth in their arguments and the groups were often failing to ponder other options and perspectives. Thus, their reflections lacked depth and criticality. Group 4 participants devoted time to considering other opinions and made compromises to reach a decision. By participant G4P2 stating that “*we have the freedom to do as we please*” indicated the group being creative and personalized the task according to their favor. For instance, G4P4 described emotions as an essential element to consider for climate action: “*Climate change can raise strong feelings such as worry fear etc. However, emotions have significant impact on action, that's true because if everyone is so shocked and they can’t do anything.”* All group 4 participants debated the relationship between action, hope and motivation. They also questioned how awareness about climate change is not enough to overcome individual consumption habits, for example being aware of the meat production process, but still choosing to eat meat in high quantity “we *could lower our consumption, but no it’s still going on...it’s rather odd... force of habit.”* They demonstrated awareness about their references, with critical perspective and ability to inquiry and answer. Thus, group 4 was able to present deep and profound reflections.

When the groups were deciding whether climate change action is a matter of individual or collective actions, all groups were able to connect their reasoning and reflections to real life issues such as “*you cannot do anything alone*” or “*there are so many structures in our society that guides the decision-making process*”. However, these groups 2, 3, 4 and 5 did not consider other aspects that could lead to action. These groups presented various levels of reflection but could not reflect on deeper level. All groups except group 1 thought that collective actions influence the most in climate actions, an example found in appendix 3. Group 1 took the time to discuss the matter from multiple perspectives before

reaching their decision. Participant G1P2 even changed her view after different aspects were considered. With their discussion to decide and make a conclusion, the group was able to properly reflect by valid arguments and justifications.

Additional outlines of reflection

Checking the time and occasionally rushing with the tasks was one pattern between the groups, as exemplified by participant G3P2 "*can we be faster now?*". All groups at least once reminded others about the time they had left. Moreover, participants from groups 1 and 3 became distracted and started to ponder the code cracking before completing the task. Other participants had to be reminded to complete the task first and not rush into the next parts. Time was another factor in task 2. All groups devoted the least time to task 2 compared to the other tasks.

Additionally, participant G4P2 demonstrated that her personal preferences were influential on her reflection process. During the game she made statements that translated her concern with sustainability. For instance, in task one, she addressed food systems, consumption and animal welfare: *"I would say maybe sustainable food and agriculture because the way we produce food right now is not sustainable […] and there's also these concerns how we treat animals, something must change*". When asked in the interview, about the outcomes of this experience, she refers to her consumption habits and how they are connected to her values: *"For me like, because I always think about the way that I consume, like buying stuff and what I eat, so now that we talked about it and those themes, I have to remind myself of my values and act with them. So, like not buying fast fashion, recycling, and stuff like that*."

4.2 Game consensus

As no model was used for assessing consensus decision-making, the results are reported in a descriptive form. Groups were able to reach consensus on their decisions quite easily. Most of the groups seemed to immediately agree with the statements that were said first, without pondering much or counterpointing it. However, tasks 1 and 4 seemed to be the ones in which people could place their arguments the most.

Although consensus seemed to have been reached easily, two different patterns were noticed among the groups. Curiously, the groups that struggled with building up their reflections demonstrated more concern in obtaining consensus. In these groups (2 and 3) the participants who were more proactive in

sharing their opinion assumed a leadership position towards consensus, whether consciously or not. These leaders however, demonstrated constant concern about other participants' opinion throughout the tasks, frequently asking *"do we agree on this?"* or simply stating *"I don't know if you all agree with that."* Despite their concern, the leaders' opinions were always prevalent, as the other participants would mainly nod in agreement or say yes. In rare cases in these groups someone counterpointed or added to the initial proposal from the leading participant. In group 2, the leading role oscillated between P1 and P2, and in group 3, P1 was more dominant. Since participant G3P1 performance outstood among the others, we decided to have a short private interview to have a better understanding of his perspective about this matter. When inquired about his leadership position in the game, he commented that his proactivity emerged after interpreting other participants' behavior and the group dynamics. "*I actually felt like nothing was happening, so I tried to make something happen*.". He added that the leading attitude is not natural to him but given the circumstances he took the role. Additionally, a participant G3P4 was also interviewed, but regarding her inactive participation. She attributed her silent role in the group dynamics to two major factors: familiarity with the participants and language. She stated:

"I didn't know that one person at all (P1), I didn't want to say my own opinion that direct if there are totally new people around […] Even though I am quite good in English, the language (affected) because I wasn't 100% sure what something means so I didn't want to say."

In Group 5, participants P1 and P3 also ensured that all participants from their group were satisfied with their decision, although this group was able to produce more quality reflections and consider all opinions. Contrary to groups 2, 3 and 5, participants from groups 1 and 4 showed less concern in asking for agreements. Everyone would deliberatively emit their opinions, which most of the time were complementary to one another.

Overall, no participant demonstrated hesitation or radical disagreement towards consensus. All participants expressed politeness and patience towards each other. All opinions were respected even on rare occasions when participants had small divergencies.

4.3 Interview results

All group members participated in the group interviews as they were conducted after the game. However, it was noticeable that as well as in the game, some people also maintained a passive role during the discussions. They mentioned that the challenges to use English language sufficiently was the major cause

of this. However, these participants listened to the conversation and joined in by nodding to some answers or using Finnish but let the rest of the participants take care of the interview. All participants from groups 1 and 4, reflected equally in the interview discussions and were able to consider different aspects and provide quality justifications for their thoughts. Given the opening of these participants the interview turned more into an open discussion in which each person was allowed to reflect on their own opinions and have a conversation about the topics.

All participants were supportive of the game and learning experience and reflected on having a "*good and fun time*" during the experiment. Groups 1 and 4 agreed on the experiment to be a comfortable way to get to know each other. They regarded it as an important feature of this game, since they would have to work with each other again in the near future, as part of the Education for the Sustainable Future course. Even though positivity was the topmost feeling, most groups also mentioned experiencing some pressure from the time and frustration when something took time, or something was confusing. Moreover, most participants mentioned language barriers to have affected the quality and quantity of their reflections. All groups were unanimous with their feelings of discussing climate change, which "*felt natural and normal*" since it is a current topic and issue to consider. Although, groups 1 and 4 were able to consider various aspects and mentioned that the constant discussion about climate change raises at the same time depressive feelings and frustration.

When the groups reflected about the game, their own learning and the opportunities of the game, the groups were unanimous that they either learned some more about climate change or it was a good reminder to think about climate change. Moreover, they thought that the game offered new ideas to teaching and the tasks helped in realizing to consider children's age when teaching about climate change. Groups 1, 4 and 5 also mentioned that the game was a fine tool to improve group dynamics and learn how to work with other people, which is essential when working as a teacher. In addition, all groups agreed that the game can influence changing perspectives if someone does not believe in climate change. Moreover, the game can help people to be aware of different perspectives, consider them and help them to know what to do. The groups unanimously agreed that the game cannot change behavior, but it can help people to start thinking about their actions.

All groups unanimously regarded consensus as important when making decisions about climate change. Groups 1, 2 and 3 added that to reach consensus it is essential to have conversations and share thoughts with others. Additionally, all groups agreed on everyone having the opportunity to share their own thoughts and join in the discussions. Although, one member from group 1 mentioned that she was not

well heard in the first task and her opinions were disregarded. Groups 1, 2, 3 and 5 agreed that reaching consensus was "*easy*" because they "*all think the same*". Participants reported that in most of the situations, someone from the groups brought up an opinion and the others automatically agreed by saying "*yes*" or nodding. So, since they shared similar opinions, there was no need to discuss it further. Group 4 indicated that they also thought the same but reached consensus only after various perspectives were considered. Groups 2 and 5 also added that they were open to various opinions and thoughts, but they were not needed since they agreed on the same opinions. Interestingly, all groups were able to mention during the interview that various opinions and aspects should be considered when discussing climate change and making decisions, but still some groups had challenges to verbally provide different opinions.

The groups generated similar answers in the interview discussions and brought out important aspects that affected their reflections. All groups were able to clearly describe the experience, express their opinions and feelings and draw conclusions, thus reflecting properly and presenting quality discussions. However, groups 1, 4 and 5 were capable of being more critical with their opinions and considering various aspects, therefore, presenting more profound reflections. All groups showed resemblance by having consensus that it was easy to work with their group and having multiple participants in the group made the tasks easier since they could complement each other. Although, participant G3P4 stated later during a one-on-one interview that she did not feel comfortable at all since one participant in the group was not familiar from before, which affected her participation. Moreover, participant G1P1 mentioned that gender could have affected their reflections. Also, another participant G4P3 mentioned that if the group had had a person who did not believe in climate change, that would have changed the whole situation and influenced group dynamics.

5. DISCUSSION

The gamified learning experience through an escape room demonstrated to be an effective tool to approach and initiate discussions about climate change. All groups were able to discuss climate change issues to different extents. Some of the tasks allowed more room for debating than others, which in turn might have affected the level of reflection of the participants. Results showed that few pre-service teachers were able to reach complete reflection about climate change according to Koole model and its indicators. Most of the participants demonstrated partial levels of reflection. Many factors could have influenced this outcome, which will be discussed subsequently. As for consensus decision-making, groups demonstrated to go through the process smoothly, easily reaching consensus. Two phenomena appeared among the groups during the process, the "leader" phenomenon and the "low reflection-high consensus" phenomenon, which will also be discussed.

RQ1. Is gamification within an escape room setup functioning as an educative tool to facilitate discussion about climate change?

Given the unprecedent reality created and faced by humans in post-modern times, as a result of the changes in climate (IPCC, 2014), it is necessary to adapt methodologies and pedagogies to engage students in meaningful education for the uncertain issues of the future (Stevenson et al., 2017). Stevenson et al. (2017) state that "Mainstreaming climate change education throughout formal education systems can be one of the most important and effective means of developing capacities for addressing the climate crisis." (p.1). Mochizuki and Bryan (2015) recognize the important role of climate change education as it has the potential for what they called a "multiplier effect". In this effect, entire communities and families can benefit once individuals share their learning experiences. The acquired knowledge can then be passed to others and eventually influence other generations, which can in turn corroborate to civic engagement and participation in decision making. As suggested by the participants, this experiment alone does not seem to be sufficient to affect individual's behavior to act more sustainably or in mitigation of climate change. However, the discussions generated during the game can be the starting point for reflection. As proposed by Koole et al. (2011), the experience is the trigger for reflection, which when done properly, passing through all the stages, generates a reflective outcome with a potential change in behavior. For pre-service teachers in particular, besides reflection

about climate change, the experience itself can generate reflection about teaching practices and pedagogies to be used in climate change education, which is useful for their future professional life.

In addition, the results corroborate the idea that discussions related to climate change and sustainability produce strong feelings such as hopelessness and frustration, as described by Cantell et al. (2019). Thus, fun climate change education activities can reduce the negative feelings around climate issues, support knowledge building and help dealing with emotions (Fabricatore & Lopez, 2012). It is also considered preferable to challenge individuals to think about solutions to problems and offer time to discuss and reflect than to constantly bombard them with devastating news and impacts of climate change, as it was highlighted in the research results and emphasized by Ouariachi and Elving (2020).

Furthermore, the results of this research point out that gamified learning situations and events can have many benefits. Climate change themed escape room was perceived by the participants as a fun method to support discussion, collaboration and to think about solutions to climate change issues, which refers to the study and conclusions by Ouariachi and Elving (2020). The method offered a chance for the participants to raise awareness about climate change issues and can remind people how much different actions and habits influence the environment. Thus, the results highlight that escape rooms can function as an educative tool to facilitate discussion about climate change. Participants were able to support their learning experiences about climate change, share their own thoughts and values and be open to new ideas and knowledge. In addition, the game allowed participants to practice and develop their educational knowledge on beneficial strategies, where climate change is implemented in teaching and learning. Even though it is known that knowledge of climate change alone is not sufficient to drive pro-environmental behavior (Kollmuss & Ayemen 2002), it is also acknowledged that lack of knowledge about the causes and consequences about climate change to be a serious barrier to mitigation and adaptation actions (Lorenzoni et al., 2007).

Games are well-known for being a good learning strategy for pupils at school level. Rowles (2012) states that games can increase cognitive and affective learning, helping to connect practice to theory. Escape rooms have been used as an effective learning strategy in adult education in different areas, such as nursing and medical schools (Brown et al., 2019). The immersive nature of escape rooms can bring participants into simulative situations in which they get to play different roles. Cantrell et al. (2017) draws attention to simulation-based learning experiences being beneficial for the development of cognitive and teamwork skills. In the current study, participants played along the storyline as if they were climate change and education decision-makers. They were able to experience the challenges of

dealing with this wicked problem, facing different tasks that mimic real-life situations under pressure. The countdown clock in the game was also an analogy of our limited time regarding acting towards climate change issues.

RQ2. How is the consensus decision-making done about climate change among pre-service teachers?

As seen, climate change is indeed a complex issue, which requires careful consideration, collaboration and thoughtful decision-making and solutions. Incropera (2015) describes climate change as a wicked problem, due to its complex nature. The author explains that such complexity is linked to different aspects of societal organization, traditions, and interests, which generates little consensus on the solutions for the issue. Regarding the consensus decision-making in this study, the results demonstrated that all groups were able to come to an agreement comfortably and perceived the process as easy. As suggested by the participants, consensus was easily reached among them because everyone was aware of climate change issues. There seemed to be consistency in agreeing on the reality and seriousness of the climate crisis. Therefore, participants' similar beliefs in climate change might have facilitated consensus. Also, despite studying different subjects, all participants were pre-service teachers, which has shown to influence their decision-making in investments for climate change mitigation. Their common background in education has also shown to be influential and relevant when the focus was educational strategies. Additionally, all participants were Finnish, which homogenizes the cultural aspects mentioned by Incropera (2005). In this study, individuals recognized the importance of pondering their own choices and actions on their own to make better decisions. The easiness on participants' decision-making process may be a consequence of the ambitious investments Finland has been doing in the past years in climate change legislation, action, and climate change education. (Jensen, 2021; Opetushallitus, 2014)

Although groups have stated such aspects in their decision-making processes, it is unclear if the consensus was reached as common agreement by every participant or if their decision was influenced. Most groups had a leader mediating the decision process but not all the leaders always asked if everyone agreed or not at the end of every task. In general, the person who expressed more specific content-knowledge inevitably took the leader position since no one would express much. Although this person showed concern about others, asking about their opinion, his/her opinion would always prevail

since the leader was the only one capable of stating something more concrete. We acknowledge that in a decision-making process there is usually a negotiation in which parties need to compromise to reach the common goal (Bressen, 2012). However, the "leader" phenomenon could have led to a negative feedback cycle, since some of the participants might have felt oppressed or intimidated by the leader's opinion, therefore not emitting their real viewpoints. This leader-follower relationship is well-known in leadership theories (Webster, 2015). The phenomenon may also be justified by conformity. In this process, people tend to be influenced by the one individual who demonstrates more confidence over a subject, so that they have a sense of belonging to the group (Sunstein & Frank, 2019). As pointed by Sunstein and Frank (2019), "If uninformed people are trying to decide whether immigration or climate change is a serious problem, […] they are likely to be responsive to the views of confident and consistent others." (p.14). Conversely, the leaders of the groups in which the leader phenomenon happened, demonstrated preoccupation in checking others' opinions. Maybe as a mechanism to minimize their leadership position and favor teamwork. Although the definition of consensus involves a group agreement, Sartor and Brown (2004) propose that each group should specify their own definitions of consensus. For certain groups that can mean everyone agreeing or disagreeing on a matter or mean that not everyone is keen on the decision, but still copes with it.

Nevertheless, it was found that all participants were considerate towards each other and valued different opinions. Such aspects are essential in consensus decision-making process, as previously discussed, and can lead to better learning outcomes (Sartor & Brown, 2004; Seeds for Change, 2010). According to Sartor and Brown (2004) consensus helps to deal with different feelings that may arise from what they call "power struggles". These conflicts of power are common between teacher and students as well as between teachers and their "superiors". In such relationships, there seems to be an advantage of the person with greater authority and victimization of the other part. However, consensus can bring the necessary understanding of the responsibilities of each of the parties, helping to overcome frustration towards a greater common goal. Thus, the ability to communicate feelings during consensus building processes may increase the trust between the parties.

Data from the interview confirmed that participation was influenced by the presence of unacquainted participants in some of the groups. Being strangers to one caused discomfort to some participants, which agree with the research of Strickland and Kaylor (2016) where a group environment can cause negative outcomes in the quality of participation. Experiencing some uncomfortable feelings increased inactiveness and influenced the group dynamics negatively. However, some groups took the game

experience as an advantage to get acquainted with one another and enhance collaboration skills. The experience strengthened the ability to increase teamworking skills, which was beneficial since all group members had to work together later during their course. Theories emphasize the importance of escape rooms to develop teambuilding and collaboration. It encourages players to communicate and find solutions to issues together as a team (Mawhirter & Garofalo, 2016; Nicholson, 2015; Veldkamp et al., 2021). Thus, these results are positive regarding the participants' future, since teamworking skills and collaboration are essential skills for a future teacher. Teachers must often work together with other colleagues to find the best solutions to support learning and establish a more functional school environment. In addition, escape rooms have shown to be a tool for groups to develop adequate conflict management skills, trust, and openness as well as consideration towards everyone's opinion (Gordon et al., 2019), which showed to appear between the participants. Those are essential features to work on when developing consensus decision-making in education as stated by Sartor and Brown (2004). Thus, the results indicate that the game enhanced education for sustainable development as such skills can be considered as building blocks of 21st century skills.

Regarding sustainability, consensus decision-making could emerge as a solution for many of the complex issues faced by humanity nowadays, in a sustainable way. Sartor and Brown (2004) claim that "we desperately need consensus decision-making processes that bring forth the creativity and intelligence of everyone in order to solve the grave problems before us" (p. 9). Classrooms are the place to begin this process, to go from a local to global scale. Therefore, pre-service teachers need to be trained to realize the importance of consensus decision-making and know how to encourage students to engage and participate in discussions with peers.

RQ3. How is the reflection process done in groups, and to what extent can pre-service teachers reflect on climate change within an educational escape room?

To Schön (1983), reflection can be a process to be developed through experience. On his "reflect-in-action" theory, he states that when one reflects critically about the situation or phenomenon, they question their own knowledge in action and then have to opportunity to fix it. The participants involved in the research experience presented various levels of reflection depending on the task. Few groups remained more consistent with their reflection levels, were able to think more critically, consider different perspectives and draw conclusions, thus, presented complete reflection by

undergoing all reflection stages. Others were presenting more superficial reflections. For Dewey (1933), reflection is a process in which each stage is determinant for the next, and the next is dependent on the previous. Therefore, it is necessary to go through all the stages for meaningful reflection. There can be multiple reasons for this variation, why not all participants were able to present all three reflection stages. Prior experiences with reflection or poor group dynamics can explain reduced quality of reflection (Koole et al., 2011; Strickland & Kaylor, 2016). Previous poor experiences or not having enough practice in reflective processes could have affected the skill levels and the willingness to discuss with others. Also, lack of focus, role taking, different values and opinions or poor interaction with other group members could have influenced the differences in reflection. For some it can be easier to work as a team, share their own thoughts and be more open-minded, which refers to Rogers (2001) who considers these points as fundamental for reflection.

Additionally, the results demonstrated that content and quantity of reflections remained scarce among some groups, which could have been influenced by participants having similar beliefs, opinions, and educational levels. It can create a situation where individuals believe that no further elaboration is needed. As Dewey (1933) proposes, reflection is grounded on beliefs, whether it is believed or disbelieved in. Beliefs are a strong component of climate action (Cantell et al., 2016). If the groups' beliefs had been more heterogenous, the discussions and consensus building could have been challenged and corresponded to the reality of stakeholders. Challenging the way of thinking and decision-making is perceived as fundamental for developing different skills such as critical thinking, problem solving and interpersonal skills (Beghetto, 2018; Bressen, 2012; Eronen & Toikka, 2021; Sartor & Brown, 2004; Seeds for Change, 2010). A challenging environment has the potential of making the experience meaningful and interesting, which was brought up in the results.

Conducting the research as a group study, where participants constantly reflected with others can be one of the explanations for mixed and superficial reflections. In their study, Seibert and Daudelin (1999) found that individual reflection was more successful and generated significantly more amounts of learning than reflection with peers. This indicates that the group environment could have influenced the outcome of reflections and the limited skills of considering various perspectives and critical thinking. While individual reflection might be more common, perceived more positive and easing the participants discussions during research experiences, Eyler et al. (1996) found group reflection equally meaningful. Those who reflect together with other people regard the experience as powerful since it can provide an opportunity to understand other perspectives, which have not been considered before.

Moreover, individuals reflecting with peers have been reported to be equally satisfied with the reflective experiences and the quality of their learning as individuals who reflect alone (Seibert & Daudelin, 1999), which also aligns with the research results. For some participants, reflection might arise after this experience, derived from the discussions reached in this experiment. Participants stated to enjoy the experience together and regarded it as meaningful. Additionally, groups offered support to each other, which was perceived as a positive experience.

Furthermore, the results demonstrated that most groups had strong individuals who were leading and contributing more to the discussions and brought more content into the reflections. Those individuals' opinions and knowledge made it easier for the groups to form their discussions and reach all stages in reflection. Such results also pointed out a possible relationship between knowledge and reflection. However, the methodology used in this study is not enough to infer if individuals who argument more indeed had more knowledge of climate change content. Dewey (1958) described the linkage between knowledge, belief and experience. “All knowing and effort to know starts from some belief, some received and asserted meaning which is a deposit of prior experience, personal and communal” (p.428). Other theories, such as the “brain-mind cycle of reflection” theory (Iran-Nejad & Gregg, 2001) proposes that self-intuitive awareness is a fundamental factor in order to reflect. As seen, reflecting on a complex problem such as climate change may be equally complex, and require better understanding of multiple factors rather than isolated reflection itself. It might be the case of exploring reflection using a more holistic approach such as systems thinking, which is highly recommended for wicked problems (Allen, 2021).

In the current study, results indicated that all the pre-service teachers who participated in the experiment had awareness of climate change, but only a few showed specific content knowledge by being able to present and discuss other perspectives. Teachers’ education about climate change is demonstrated to be a growing concern around the world. Many countries have assessed the level of knowledge and understanding of higher education students, particularly pre-service teachers. Research conducted in the US, Canada and Australia have found that pre-service teachers had limited knowledge of climate change, even those studying science-related subjects (Boon, 2016). Furthermore, Boon (2016) investigated pre-service teachers’ understandings of climate change, based on the premises that effective pre-service teacher education has the potential to prepare citizens to act for mitigation and adaptation of climate change. She found that pre-service teachers presented a consistent engagement in climate change matters and some of them obtain their knowledge of climate change

from practical activities, including group projects. Thus, it should be seen as essential to prepare pre-service teachers for their future profession and develop their understanding of climate change, providing them with pedagogies that go beyond the delivery of knowledge and that allow them to be creative in their future pedagogies too.

In addition to the variability of reflections, the results proved language skills to be a barrier for the quality of reflection among a few participants. Some had challenges in committing to the tasks and discussions since they were not so familiar with English vocabulary and comfortable with using English. While challenging and uncomfortable situations are perceived to develop reflection skills, some participants were not able to receive support from their mother tongue to reflect, which refers to Rogers (2001) who highlights the importance of support systems. The results demonstrated that participation and reflection could have been higher quality if participants had been able to use Finnish, their mother tongue. Furthermore, as stated in the results, the pressure from the time could have limited participants abilities and quality to reflect. Therefore, it can be concluded that most members in the groups participated in the reflection processes, yet some individuals influenced the reflection outcomes more. Reflection occurred both during action and after the experience to increase the reflective outcomes, which is considered as beneficial (Rogers, 2001). Moreover, gamification can be an effective tool to promote individual reflection about climate change to some extent. While most participants proved to consider essential elements and justify their thoughts, for others the environmental effects and factors influenced negatively in their reflections.

The research results raised important aspects which are considered in the Finnish national core curriculum (Opetushallitus, 2014). Participants were able to see the importance of incorporating sustainable lifestyles into individuals' own lives and including families as well. The group's answers indicated that they perceive school and home collaboration as essential since it was suggested that school and home can work together to develop and enhance sustainable actions. Incorporating families in the school environment and collaborating with home is seen as essential in Finnish school according to the national core curriculum. It is believed that continuous collaboration can influence children’s mindset and values and support growth even better. Furthermore, the game tasks were developed by aligning with the national core curriculum. Reflection and the ability to describe own learning processes are believed to be essential factors to support learning and especially life-long learning (Koole et al., 2011). Life-long learning is valued in Finnish education, and it attempts to equip all students with tools for life-long learning (Opetushallitus, 2014). The research was aiming to encourage

participants to reflect, see the value of it and continue to improve their own skills. Moreover, the research experience offered a chance to the participants to develop their skills in assessing news and practice how to be critical towards given information. This was important practice since Finnish education demonstrates importance in educating people against fake news. The national core curriculum (2014) describes the significance of teaching students how to be critical towards news and information and how to search knowledge and facts responsibly. These points are also considered in climate change education, which is perceived as a tool against fake news (Boon, 2016). Once people are equipped with knowledge, they can recognize elements of fake news and real articles. After easy recognition, groups reflection was limited because of the obviousness of these elements. The task was too objective. To stimulate reflection, the task could have been elaborated in a way that encouraged the participants to think of the implications of these articles in climate change education.

Given the importance and the complexity of climate change, individuals and future generations should be educated to have the knowledge to know how to respond to the issues. Therefore, teachers must be equipped with the ability to share knowledge and provide effective climate change education. In such a task, teachers should be able to welcome different perspectives whilst helping to review the misconceptions that may rise on the subject, that can be supported by diverse socio-cultural factors. As education for sustainable development is aiming at responding to issues that transcend and connect social, environmental, and economic aspects, innovative pedagogies should be implemented to foster collaboration, critical and creative thinking as well as capacity building (Stevenson et al., 2017). This way youth can be prepared to meet the enormous expectations that were laid on them regarding a sustainable future. As emphasized by Jickling (2013) there is an expectation that children at school should be empowered to adapt to the future, even though the current generation seems uncapable to do so. Lastly, reflection appears to have an important relation with consensus decision-making. Following Dewey's reflective thinking model, makes it easier to understand how stages of both processes overlap and are complementary to each other. In both processes it is necessary to first identify the problem, then propose solutions for the problem in question, while considering multiple perspectives and evidence that help anticipate consequences. Both processes are essential for teachers and demonstrated to be powerful tools to be used in climate change education.

6. CONCLUSION AND IMPLICATIONS

The study examined pre-service teachers' reflection and consensus decision-making processes about climate change in an educational escape room. From the findings, it can be concluded that reflection and consensus decision-making are closely tied together. For a group to be able to build consensus over a subject and come to a decision together as a team, all members should reflect their opinions and thoughts. Careful consideration of different perspectives and argumentation are needed to strive for consensus towards a common goal. Furthermore, reflection and consensus are essential in battling against climate change and to reduce distortion around it. There is a need to assess individuals' own previous experiences regarding climate change, and critically reflect on new perspectives and behavior to enable effective action. Such steps are just a small part, but rather fundamental in tackling climate change issues. Transferring these processes to education can be challenging and an enormous responsibility, since we are all learning together through the journey. Climate change is indeed a wicked problem, and as an analogy of the game, we need to invest "all of our coins" in it to try to find the right responses in short term, because "the clock is ticking". To do so, we need to promote innovative pedagogies that align with the demands of the new generation of learners while stimulating the development of essential skills.

This experiential research proposed the use of a gamified learning space to assess how pre-service teachers could reflect and make decisions over such delicate and complex issues. Assessing group reflection in these circumstances proved to be challenging, particularly considering its subjectiveness and the multiple factors that may have influenced the participants. The results still showed that the research groups were presenting various reflection levels, in some cases certain individuals increasing the quality of reflections, finding consensus at times, and making decisions as a group. However, pre-service teachers proved to need more practice in developing reflection and consensus decision-making skills. Both skills have powerful influences on various other important skills and factors that are essential in their professional life, such as life-long learning, interpersonal skills, knowledge, and sustainable education. Therefore, it is strongly advised that both skills are continuously trained. With regular practice pre-service teachers can increase their abilities and language skills, thus, become confident reflective practitioners and decision-makers. Nonetheless, pre-service teachers need to be supported in their education as well. There should be opportunities for them to experience experiential learning situations that will allow them to reflect and implement the outcomes of such reflection into their teaching practices.

Beyond reflection, this escape room was also found to be an effective tool to debate climate change. Participant groups were able to raise valid points, exposing previous personal experiences, feelings, and opinions, from where they were able to elaborate decisions together. Although external factors such as language skills, natural pressure from the environment and peers influenced the quality of reflections and decisions, they were still able to reach the goal of this research. Even though the experience did not influence participants' behavior towards climate change it was successful in affecting their opinions and thoughts to some extent. The fun aspect of the game demonstrated how to have a positive impact on the learning experience, which was new for some of the participants. As a side effect, the experience showed to improve group cohesion and teamwork. We hope that this success increases curiosity to ponder climate change issues and especially our own actions in the future as a teacher and an individual.

7. FUTURE RECOMMENDATIONS

To further the research, it would be beneficial to include international pre-service teachers to increase the validity of the results. The heterogeneity of backgrounds supported by different beliefs and traditions could enrich the discussions bringing different perspectives into play. Another possibility would be to offer a chance for the groups to participate in the game by using their mother tongue to further investigate if it affects the results. The results could reveal whether language skills were an actual barrier to the reflection quality. An additional strategy would be testing different group compositions, such as “well-integrated” versus “strangers-to-each-other” or delimitating pre-service teachers within the same subject. Being mindful of group composition could help isolate the variables to find the effects on the discussion, reflection levels and whether groups would have challenges in building consensus for to make decisions. A different approach could include assessing knowledge, beliefs and values and their correlation to reflection. Assessing reflection turned out to be challenging since various factors may have influenced the reflection and consensus quality. Thus, the aforementioned points could be essential to improve the research even further.

The escape room as a learning environment can provide infinite possibilities in qualitative and quantitative research. In terms of quantitative research knowledge of climate change and climate change education could be assessed, which is a more common approach to the use of climate change related escape rooms (Ouariachi & Elving, 2020). On the qualitative aspect, a thorough examination of emotions and an individual assessment of reflection could be more appropriate to confirm if this one-time experience is sufficient to evoke reflection about climate change.

References

Adams, V., Burger, S., Crawford, K., & Setter, R. (2018). Can You Escape? Creating an Escape Room to Facilitate Active Learning. *Journal for Nurses in Professional Development, 34*(2), 1-5. DOI: 10.1097/NND.0000000000000433.

Allen, W. (2021, November 7). An introduction to systems thinking and systemic design – concepts and tools - Learning for Sustainability. Learning for Sustainability. https://learningforsustainability.net/post/systemicdesign-intro/

Ash, S.L., & Clayton, P.H. (2004). The Articulated Learning: An Approach to Guided Reflection and Assessment. *Innovative Higher Education, 29*(2), 137-154.

Beghetto, R. A. (2018). What if?: building students' problem-solving skills through complex challenges. ASCD.

Bernardo, C., Wang, L., Vasca, F., Hong, Y., Shi, G., & Altafini, C. (2021). Achieving consensus in multilateral international negotiations: The case study of the 2015 Paris Agreement on climate change. *Science Advances, 7*(51). https://doi.org/10.1126/sciadv.abg8068

Blaxter, L., Hughes, C., & Tight, M. (2010). *How to research (4th ed.)*. Maidenhead, England: McGraw-Hill/Open University Press.

Bressen, K.T. (2012). Consensus Decision-Making: What, Why, How. In Orsi, J. (2012). *Practicing law in the sharing economy: helping people build cooperatives, social enterprise, and local sustainable economies*. Chicago, Illinois: American Bar Association.

Brown, N., Darby, W., & Coronel, H. (2019). An escape room as a simulation teaching strategy. *Clinical Simulation in Nursing, 30*, 1-6.

Cantell, H., Tolppanen, S., Aarnio-Linnanvuori, E., & Lehtonen, A. (2019). Bicycle model on climate change education: presenting and evaluating a model. *Environmental Education Research, 25*(5), 717-731, DOI: 10.1080/13504622.2019.1570487

Cantrell, M., Franklin, A., Leighton K., Carlson, A. (2017) The evidence in simulation-based learning experiences in nursing education and practice: An umbrella review. *Clinical Simulation in Nursing, 13* (2017), pp. 634-667, 10.1016/j.ecns.2017.08.004

Caponetto, I., Earp, J., & Ott, M. (2014). Gamification and education: A literature review. *European Conference on Games Based Learning, 1,* 50-57.

Clarà, M. (2015). What Is Reflection? Looking for Clarity in an Ambiguous Notion. *Journal of Teacher Education, 66*(3), 261–271.

Clarke, S., Peel, D., Arnab, S., Morini, L., Keegan, H., & Wood, O. (2017). EscapED: A Framework for Creating Educational Escape Rooms and Interactive Games for Higher/Further Education. *International Journal of Serious Games, 4*(3), 73-86. 10.17083/ijsg.v4i3.180.

Climate University (n.d.) Climate.now [MOOC]. Digicampus. https://digicampus.fi/course/view.php?id=3548

Cohen, L., Manion, L., & Morrison, K. (2007). *Research Methods in Education* (6th edition). Routledge.

Cowie, J. (2013). *Climate change: biological and human aspects (2nd ed.)*. Cambridge [England]: Cambridge University Press.

Dewey, J. (1933). *How We Think: A Restatement of the Relation of Reflective Thinking to the Educative Process.* Houghton Mifflin.

Dewey, J. (1958). *Experience and Nature*. Courier Corporation

Dillon, J., Stevenson, R., & Wals, A. (2016). Introduction to special section: Moving from citizen to civic science to address wicked conservation problems. *Conservation Biology, 30*(3), 450–455.

Eronen, L., & Toikka, S. (2021). Alkuopetusikäisen valmius reflektoida matemaattisessa ongelmanratkaisutilanteessa. *FMSERA Journal, 4*(1), 1–15. Retrieved from https://journal.fi/fmsera/article/view/95475

Eyler, J., Giles, D.E., & Schmiede, A. (1996). *A practitioner's guide to reflection in service-learning: Student voices and reflections*. Nashville, TN: Vanderbilt University.

Eskola, J., & Suoranta, J. (1998). *Johdatus laadulliseen tutkimukseen.* Vastapaino.

Fabricatore, C., & Lopez, X. (2012). Sustainability Learning through Gaming: An Exploratory Study. *Electronic Journal of e-Learning, 10*(2), 209-222.

Freire, P. (1993). *Pedagogy of the oppressed*. New York: Continuum.

Füssel, H. M. (2007). Adaptation planning for climate change: concepts, assessment approaches, and key lessons. *Sustainability science, 2*(2), 265–275.

García, F. J. L., Kevany, K., & Huisingh, D. (2006). Sustainability in higher education: what is happening?. Journal of Cleaner Production, 14(9-11), 757-760.

Golafshani, Nahid. (2003). Understanding Reliability and Validity in Qualitative Research. *The Qualitative Report, 8*(4), 597-607. 10.46743/2160-3715/2003.1870.

Gordon, S. K., Trovinger, S. N., & DeLellis, T. (2019). Escape from the usual: Development and implementation of an "escape room" activity to assess team dynamics. Currents in Pharmacy Teaching and Learning, 11(8), 818–824. https://doi.org/10.1016/j.cptl.2019.04.013

Greetham, B. (2014). *How to write your undergraduate dissertation (2nd edition)*. Palgrave.

Gupta, J. (2010). A history of international climate change policy. *Wiley Interdisciplinary Reviews: Climate Change, 1*(5), 636-653.

Hamari, J., Koivisto, J., & Sarsa, H. (2014). Does Gamification Work? – A Literature Review of Empirical Studies on Gamification. In *proceedings of the 47th Hawaii International Conference on System Sciences*, Hawaii, USA, January 6-9, 2014.

Huotari, K., & Hamari, J. (2012). Defining gamification - A service marketing perspective. In *Proceedings of the 16th International Academic MindTrek Conference 2012: "Envisioning Future Media Environments", MindTrek 2012* (pp.17-22) https://doi.org/10.1145/2393132.2393137.

Incropera, F. P. 2015. Climate Change: A Wicked Problem – Complexity and Uncertainly at the Intersection of Science, Economics, Politics and Human Behaviour. Cambridge: Cambridge University Press

IPCC, 2014: Climate Change 2014: Mitigation of Climate Change. Contribution of Working Group III to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change [Edenhofer, O., R. Pichs-Madruga, Y. Sokona, E. Farahani, S. Kadner, K. Seyboth, A. Adler, I. Baum, S. Brunner, P. Eickemeier, B. Kriemann, J. Savolainen, S. Schlömer, C. von Stechow, T. Zwickel and J.C. Minx (eds.)]. Cambridge University Press, Cambridge, United Kingdom and New York, NY, USA.

Jagersma, J. (2010). Empowering Students as Active Participants in Curriculum Design and Implementation. In *ERIC* (No. ED514196). Retrieved March 8, 2023, from https://eric.ed.gov/?id=ED514196

Jensen, L. (2021). *EU progress on climate action – How are the Member States doing? Climate action in Finland: Latest state of play* (PE 696.187). European Parliamentary Research Service. https://www.europarl.europa.eu/RegData/etudes/BRIE/2021/696187/EPRS_BRI(2021)696187_EN.pdf

Jickling, B. (2013). Normalizing catastrophe: An educational response. *Environmental Education Research, 19*(2), 161–176.http://dx.doi.org/10.1080/13504622.2012.721114

King, G., Keohane. R. O., & Verba, S. (1994). *Designing Social Inquiry: Scientific Inference in Qualitative Research.* Princeton University Press.

Koole, S., Dornan, T., Aper, L., Scherpbier. A., Valcke. M., Cohen-Schotanus. J., & Derese. A. (2011). Factors confounding the assessment of reflection: a critical review. *BMC Medical Education, 11*(104). https://doi.org/10.1186/1472-6920-11-104

Kollmuss, A., & Agyeman, J. (2002). Mind the gap: why do people act environmentally and what are the barriers to pro-environmental behavior? *Environmental education research, 8*(3), 239-260. http://dx.doi.org/10.1080/13504620220145401

Lee, J., & Hammer, J. (2011). Gamification in Education: What, How, Why Bother? *Academic Exchange Quarterly 15*(2). 1-5.

Li, M-C., & Tsai, C-C. (2013). Game-Based Learning in Science Education: A Review of Relevant Research. *Journal of Science Education Technology, 22*, 877–898. https://doi-org.ezproxy.uef.fi:2443/10.1007/s10956-013-9436-x

Loizos, P. (2000). *Video, film and photographs as research documents*. In Bauer, M. W., & Gaskell, G. (Eds.), *Qualitative researching with text, image and sound* (pp. 94-107). SAGE Publications. https://www-doi-org.ezproxy.uef.fi:2443/10.4135/9781849209731

López-Pernas, S., Gordillo, A., Barra, E., & Quemada, J. (2019). Examining the Use of an Educational Escape Room for Teaching Programming in a Higher Education Setting. *IEEE Access, 7*, 31723-31737. 10.1109/ACCESS.2019.2902976.

Lorenzoni, I., Nicholson-Cole, S. & Whitmarsh, L. (2007). Barriers perceived to engaging with climate change among the UK public and their policy implications. *Glob Environ Change 17*(3), 445-459. http://dx.doi.org/10.1016/j.gloenvcha.2007.01.004

Mac Gregor, M. (2018). Campus Clue: Habituating Students to the Information Search Process via Gaming. *Pennsylvania Libraries: Research & Practice, 6*(2), 86-92. https://doi.org/10.5195/palrap.2018.172

Martella, R. C., Nelson, J. R., Morgan, R. L., Marchand-Martella, N. E., & Marchand-Martella, N. E. (2013). *Understanding and interpreting educational research*. Guilford Publications.

Mawhirter, D., & Garofalo, P. (2016). Expect the unexpected: Simulation games as a teaching strategy. *Clinical Simulation in Nursing, 12*(4), 132-136. doi:10.1016/j.ecns.2015.12.009

Mochizuki, Y., & Bryan, A. (2015). Climate change education in the context of education for sustainable development: Rationale and principles. *Journal of Education for Sustainable Development, 9*(1), 4-26.

Moon, J. A. (1999). Reflection in learning and professional development: Theory and practice. Kogan Page. https://doi.org/10.4324/9780203822296

Moura, A., & Santos, I. L. (2020). Escape room in Education: Gamify Learning to Engage Students and Learn Maths and Languages. In Bento, M., & Lencastre, J. (Eds.) (2019). *Experiences and perceptions of pedagogical practices with Game-Based Learning & Gamification* (p. 179-193). Research Centre on Education (CIEd).

Nicholson, S. (2015). *Peeking behind the locked door: A survey of escape room facilities.* http://scottnicholson.com/pubs/erfacwhite.pdf.

Opetushallitus. (2014). Perusopetuksen opetussuunnitelman perusteet 2014. Helsinki: Opetushallitus.

Osman, K., Radzi, S.N.F., & Mohd Said, M.N. (2022, May 23). Climate Change Education. In Encyclopedia. https://encyclopedia.pub/entry/23226

Ouariachi. T., & Elving. W. J.L. (2020). Escape rooms as tools for climate change education: an exploration of initiatives. *Environmental Education Research, 26*(8),1193-1206, DOI: 10.1080/13504622.2020.1753659

Piñero Charlo, J.C., Belova, N., Quevedo Gutiérrez, E., Zapatera Llinares, A., Arboleya-García, E., Swacha, J., López-Serentill, P., & Carmona-Medeiro, E. (2022). Preface for the Special Issue “Trends in Educational Gamification: Challenges and Learning Opportunities”. *Educ. Sci., 12*, 179. https://doi.org/10.3390/educsci12030179

Radzi, S. N. F., Osman, K., & Mohd Said, M. N. (2022). Progressing towards Global Citizenship and a Sustainable Nation: Pillars of Climate Change Education and Actions. *Sustainability, 14*(9), 5163. https://doi.org/10.3390/su14095163

Rapley, T. (2004). *Interviews*. In Seale, C., Gobo, G., Gubrium, J. F., & Silverman, D. (Eds.), *Qualitative research practice* (pp. 16-34). SAGE Publications. https://www-doi-org.ezproxy.uef.fi:2443/10.4135/9781848608191

Ratinen, I. (2011). Primary Student-Teachers' Conceptual Understanding of the Greenhouse Effect: A Mixed Method Study. *International Journal of Science Education, 35*(6),1-27. DOI:10.1080/09500693.2011.587845.

Rinaldi, C. (2016). *The Hundred Languages in Ministories: Told by the Teachers and Children from Reggio Emilia.* Davis Publications.

Rogers, R. R. (2001). Reflection in Higher Education: A Concept Analysis. *Innovative Higher Education, 26*(1), 37-57.

Rowles, C.J. (2012). Strategies to promote critical thinking and active learning. In: Billings,D.M., Halstead, J.A. (Eds.), Teaching in Nursing: A Guide for Faculty, fourth ed. Elsevier, St. Louis, MO, pp. 258–284.

Sartor, L., & Brown, M. Y. (2004). *Consensus in the Classroom*. Amsterdam University Press.

Schön, D. A. (1983). *The Reflective Practitioner: How Professionals Think in Action*. Routledge.

Seeds for Change. (2010). Consensus decision making: A guide to collaborative decision-making for activist groups, co-ops and communities. www.seedsforchange.org.uk

Seibert, K. W., & Daudelin, M. W. (1999). *The role of reflection in managerial learning: theory, research, and practice*. Westport, CT: Quorum.

Siegle, D. (2015). Technology: Learning Can Be Fun and Games. *Gifted Child Today, 38*(3), 192–197. https://doi.org/10.1177/1076217515583744

Smiderle, R., Rigo, S., Marques, L., Coelho, J., & Jaques, P. (2020). The impact of gamification on students' learning, engagement and behavior based on their personality traits. *Smart Learning Environments, 7*(3). 10.1186/s40561-019-0098-x.

Stevenson, R.B., Nicholls, J. & Whitehouse, H. What Is Climate Change Education?. Curric Perspect 37, 67–71 (2017). https://doi.org/10.1007/s41297-017-0015-9

Sunstein, C. R., & Frank, R. H. (2019). *Conformity: The Power of Social Influences*. New York: NYU Press. https://doi.org/10.18574/9781479896585

Sorenson, R. P. (2011). *Eunice Foote's pioneering research on CO2 and climate warming*. Search and Discovery.

Strickland, H. P., & Kaylor, S. K. (2016). Bringing your a-game: Educational gaming for student success. *Nurse Education Today, 40*, 101-103. https://doi.org/10.1016/j.nedt.2016.02.014

Taylor, S. J., Bogdan, R., & DeVault, M. (2016). *Introduction to Qualitative Research Methods: A Guidebook and Resource*, 4th edition. John Wiley & Sons, Incorporated.

United Nations General Assembly, Transforming our world: the 2030 Agenda for Sustainable Development, 21 October 2015, A/RES/70/1. https://www.refworld.org/docid/57b6e3e44.html Retrieved 31 January 2023.

UNESCO. (2019). *ESD and GCED Up Close: Cognitive, Social and Emotional, and Behavioural Learning in Education for Sustainable Development and Global Citizenship from Pre-Primary to Secondary Education*. Paris: UNESCO.

UNESCO. (2020). Education for Sustainable Development: A Roadmap. UNESCO Publishing.

UNESCO. (2022). Youth demands for quality climate change education (ED-2022/WS/33). Retrieved March 14, 2023, from https://unesdoc.unesco.org/ark:/48223/pf0000383615

United Nations Framework Convention on Climate Change. (2022, November 20). *COP27 Reaches Breakthrough Agreement on New "Loss and Damage" Fund for Vulnerable Countries* [Press release]. https://unfccc.int/news/cop27-reaches-breakthrough-agreement-on-new-loss-and-damage-fund-for-vulnerable-countries

Veldkamp, A., Knippels, M-C. P. J., & van Joolingen, W. R. (2021). Beyond the Early Adopters: Escape Rooms in Science Education. *Front. Educ, 6*. doi: 10.3389/feduc.2021.622860

Vesa, M. (2021). *Organizational Gamification: Theories and Practices of Ludified Work in Late Modernity*. Routledge.

Webster, V. J. (2015). *The Dark Side of Leadership and Its Impact on Followers* [PhD dissertation]. Griffith University. Doi: https://doi.org/10.25904/1912/1567.

Appendices

Appendix 1: interview questions

- How did you feel about participating in this escape room?
- How do you feel about discussing climate change?
- How relevant do you think consensus is in climate change issues?
- How was it for you to make decisions in a group?
- Can you describe the process of coming to an agreement in your group?
- What do you think is your learning outcome from this experience?
- Do you think the experience has the potential to change perspectives about climate change?
- Do you think the experience has the potential to change behavior towards climate change?
- How do you think you can use this experience to develop as a teacher?

Appendix 2: Tasks

Task 1. The investment

Instructions. Your task as representatives is to negotiate with stakeholders for funding to mitigate climate change effects by **2025**.

- You will find 7 flashcards with main topic areas for mitigating climate change.
- You will also find 3 coins that will be the symbol of your investment money.

As a group you need to choose 3 topic areas, which you want to invest your money in. Discuss and explain why you decided to pick those 3 areas. Be specific with your answers. Place the coins on top of the flashcards you choose. Using the tablet, take a picture of the final decision.

Topic areas for mitigating climate change (cards)

- Education
- Energy security and Renewable energies
- Affordable and Sustainable Transportation and Mobility
- Improved air quality and health outcomes
- Adaptation Measures to Climate Disaster
- Sustainable Food and Agricultural Systems
- Gender Equality Actions

Hint: To escape this room you will have to crack the code and open the door. The clock is ticking, the issues don't wait. Four numbers it has. The decisions made now will impact the near future.

Press: number#

Answer: 2025

Task 2. Fake or Fact

Instructions: Climate change crisis has become a hot topic in media. There is scientific evidence about climate change, but there is still controversy about the topic, which is supported by fake news. Read the following news and debate, whether they are fake or **fact**.

News 1. Fact

Evidence is now 'unequivocal' that humans are causing global warming – UN report. Retrieved from: https://perma.cc/LEW6-6PEK

News 2. Fake

Climate change is an obvious myth – how much more evidence do you need? Retrieved from: https://www.theguardian.com/science/brain-flapping/2014/nov/25/climate-change-is-an-obvious-myth-how-much-more-evidence-do-you-need

Hint: By debating you can unlock many things, such as different beliefs and points of view. But here you need to unlock padlocks. Crack the code to unlock your way to the next task.

Puzzle to crack the code:

A= 3	**H= 8**	**O= 6**	**V= 3**
B= 8	**I= 3**	**P= 1**	**X= 6**
C = 1	**J= 5**	**Q= 4**	**Y= 1**
D= 6	**K= 9**	**R= 8**	**Z= 7**
E= 7	**L= 1**	**S= 9**	
F= 2	**M= 2**	**T= 2**	
G= 5	**N= 4**	**U= 5**	

Answer: FACT = 2312

Task 3. Curriculum Planning

Instructions: You were selected to be part of the National Educational Board. Read the quotes from the Finnish National Core Curriculum in the flashcards (Finnish National Board of Education, 2016). Describe which strategies would you use to integrate climate change education in a in a primary school level lesson. Debate the reasons why certain strategies could be more effective than others. At the end, list the strategies that you (as a group) think would be more effective.

Underlying values of basic education

“Ecosocial knowledge and ability means that the pupils understand the seriousness of climate change, in particular, and strive for sustainability” (p.16)

Key Content Areas Related to the objectives of Environmental Studies in grades 3-6

“Aspects to be considered in the selection of contents include fostering biodiversity, climate change and its mitigation, sustainable use of natural resources, promotion of health, caring for the pupil’s own cultural heritage, living in a multicultural world and the global welfare of the humankind now and in the future. They practice environmentally responsible actions...and participation and involvement at the local or global level.” (p.260)

Hint: There are many keys to tackle climate change, but you only need one to escape the room. Solve the riddle and find your way out. Has 88 keys, but not one of them will open a door.

Answer: Piano. (Use the key in the envelope to open the piano keyboard, there they will find another key to open the lock at the door)

Task 4. Evaluating values

Instructions: You gather together as specialists to understand what factors influence pro-climate action. You will find 2 envelopes.

1. contains: 7 factor cards

2. contains: 2 final decision cards

- As a group you should debate and organize the 7 factor cards, ranking them in order of what you judge to be more impactful to climate change action.

- At the end, using the final decision cards, decide as a group if climate change is a matter of individual or collective actions. Place your final decision on the marked area.

Cards

Knowledge & thinking skills Understanding certain aspects involved in climate change issues and their impact on the environment and life. Also, knowledge of the possibilities and limitations of technical solutions. Understanding the importance of reflection, examining and questioning.	**Identity, values, worldview** Looking environmental issues from different perspectives. What individuals consider to be important. Identity, values and worldview shape the perception of climate impacts and adaptation to them.
Action Action refers to efforts made to control and mitigate climate change. Environmental action is related to not only knowledge of a topic, but also empathy, a desire to help, other people's opinions, societal norms and ease of action.	**Motivation & Participation** Action requires motivation to act. Other's opinions can maintain and increase motivation. Participating in action together with others can support involvement.
Operational barriers Barriers to environmental responsibility are often human tendencies such as desire for comfort, laziness, force of habit and haste. Lack of environmental knowledge, psychological and socio-cultural barriers, such as denying the problem, social norms and uncertainty about the effectiveness of one's own actions can have an effect.	**Hope & other emotions** Climate change can raise strong feelings, such as worry, fear, sadness, guilt, hatred and hopelessness. Also, other's negative attitudes and feelings can have an effect on attitudes and capacity-building. However, emotions have significant impact on action. Feeling strongly about a certain issue can increase action. Sharing ideas and searching for answers to challenging questions brings hope.
Future orientation Envisioning the future and thinking critically about the future.	**Collective** Global climate change has become the collective action problem of our era. If current international efforts are insufficient to address the rising emissions of green- house gases (GHGs), a changing climate will significantly affect the entire planet and its inhabitants, human and nonhuman

Individual Our lifestyles have a profound impact on our planet. Our choices matter. Around two-thirds of global greenhouse gas emissions are linked to private households. The energy, food, and transport sectors each contribute about 20 per cent of lifestyle emissions. From the electricity we use, to the food we eat and the way we travel, we can make a difference.	

Hint: To escape the game, solve the mystery. "Sometimes it is hard to see what is right in front of you. What can the numbers behind the cards do together?"

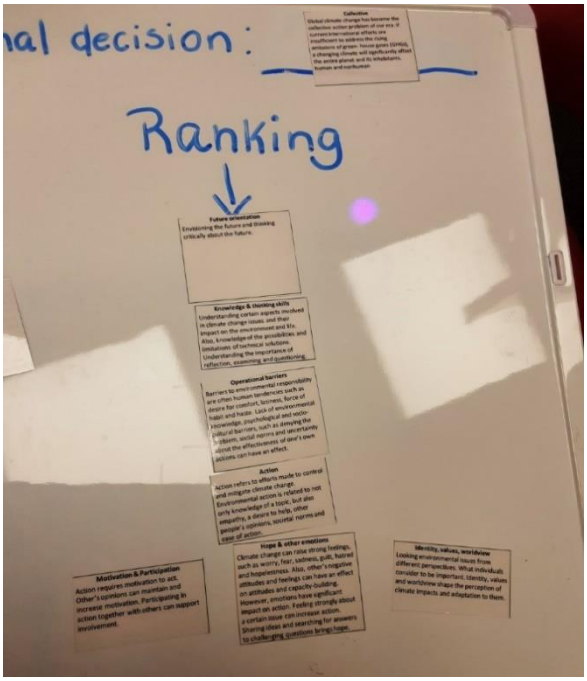
Answer: there are numbers behind the cards that light up using an uv flashlight. They need to sum up the numbers to breakout the room.

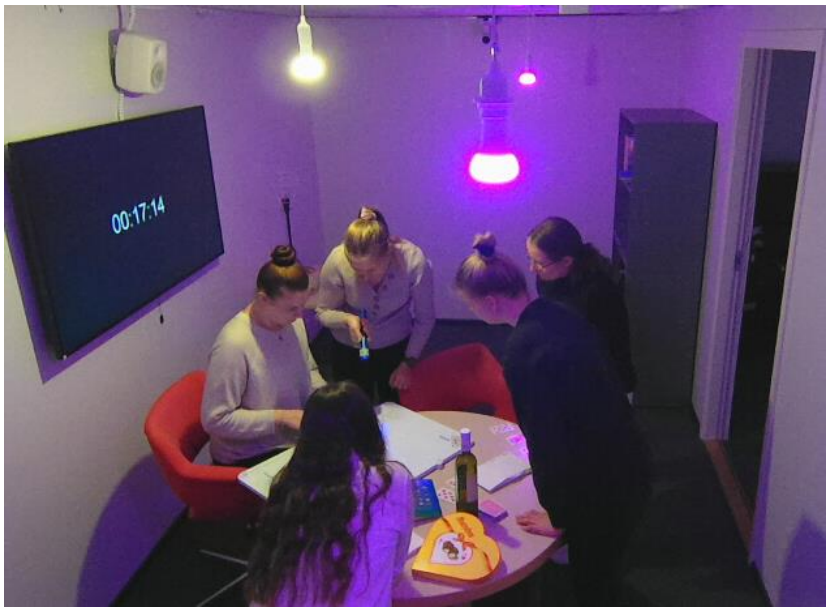
Appendix 3: Game pictures

One of the groups' resolution of task 1. The coins symbolize money to invest in each of the areas. This group prioritized education, energy security and renewable energy and sustainable food and agricultural system.



One of the groups' resolutions of task 4. The group concluded as a final decision that climate change is a matter of collective action.




00:17:14